A Theory of Endogenous Aspirations

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Abstract

People choose aspirations that influence how they evaluate outcomes and the actions they take. But those actions can also determine which aspirations are worth choosing. This paper develops a model in which an agent jointly chooses an action and a feasible aspiration before uncertainty resolves. A higher aspiration can be more rewarding when it is attained, but reduces by how much a given outcome exceeds it and increases the cost of falling short. Here, loss sensitivity scales the penalty for falling below the chosen aspiration. For finite lotteries, there is a cutoff in loss sensitivity: below it, the agent chooses an aspirational reference point that can be missed; above it, a no-loss reference point attained in every state. In binary lotteries, a higher probability of the high outcome or greater payoff dispersion can raise this cutoff, so the same individual may aspire in one environment but not another. I then allow the action to change the lottery and compare how the agent responds to incentives when her aspiration adjusts and when it is held fixed. In portfolio choice, allowing wealth targets to adjust amplifies the shift toward the state whose payoff becomes relatively cheaper and, in the aspirational regime, reinforces the shift toward insurance as loss sensitivity increases. In effort choice, greater loss sensitivity increases effort when a fixed aspiration can be missed, but can instead reduce effort at intermediate productivity when aspirations adjust. Finally, although aspirations are not observed, their optimality across lotteries allows preference parameters to be identified from choice data.

Keywords: endogenous reference points; goals; aspirations; loss sensitivity; loss aversion; risk taking.

JEL codes: D01; D81; D91; G11.

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1 Introduction

People often set goals or hold aspirations that guide their decisions. In education, students set academic goals that can affect how much they study and how well they perform (Clark et al., 2020). In finance, wealth targets have long been used to explain risky choice and portfolio allocation (Payne et al., 1980; Shefrin and Statman, 2000; Das et al., 2010). More generally, the aspirations literature has studied how aspirations respond to social and economic circumstances (Genicot and Ray, 2017; Dalton et al., 2016), but individuals can also have some agency over the aspirations they set. Setting those aspirations involves a trade-off: a more ambitious aspiration can be more rewarding when it is attained, but it can increase the chance and cost of falling short. The problem becomes more complex when individuals can also take actions, such as exerting effort, that change the distribution of outcomes. Aspirations affect the incentives to take an action, while that action also determines which aspirations are worth setting. This paper studies how people choose their aspirations and how these choices influence decisions under uncertainty.

Aspirations can serve as reference points because they determine whether realized outcomes are perceived as successes or failures (Heath et al., 1999). Psychological research also finds that people expect more satisfaction from attaining a harder goal, and that their satisfaction depends on how far performance exceeds or falls short of the goal (Meyer et al., 1988; Locke and Latham, 2002). Reference dependence helps explain how homeowners set asking prices, how unemployed workers search for jobs, and how much effort people exert.[1] Where the reference point lies therefore matters, and so does how it is formed. Much of the reference-dependence literature either takes the reference point as given or assumes that it is supplied by the decision environment.[2] In expectations-based models (Kőszegi and Rabin, 2006, 2007), the reference point is determined by expectations over outcomes, which must be consistent with the agent's choice. I instead allow the agent to choose the reference point itself.

I develop a model in which an agent jointly chooses an action and a feasible reference point, before uncertainty resolves, to maximize ex-ante utility. I first characterize how the agent chooses a reference point for a given risky prospect and how this choice varies with preferences and economic opportunities. Next, I address a well-known challenge in the reference-dependence literature: reference points are typically unobserved, making it difficult to distinguish variation in preferences from variation in the reference point itself. Under a specification in which the utility components share a curvature parameter, I show that preference parameters can nonetheless be identified, because the unobserved reference point must be optimal for every lottery the agent evaluates. Finally, I study how reference-point adjustment affects behavior in applications to portfolio choice and effort

[1] See Genesove and Mayer (2001), DellaVigna et al. (2017), and Abeler et al. (2011).

[2] See Kahneman and Tversky (1979); Tversky and Kahneman (1991); Masatlioglu and Ok (2014); Alaoui and Penta (2026).

provision. In portfolio choice, when a state becomes relatively cheaper, the investor shifts more wealth toward it when her wealth target adjusts than when the target is fixed. In effort choice, a higher cost of falling short increases effort when aspirations are fixed, but can instead reduce effort when aspirations adjust.

The agent faces a menu of finite lotteries. Before uncertainty resolves, she chooses a lottery from the menu and a reference point within the range of outcomes available in the menu. An outcome is then drawn according to the chosen lottery, and its utility depends on whether it reaches the reference point. When the outcome reaches the reference point, utility depends on both the level of the reference point and the amount by which the outcome exceeds it. When the outcome falls below the reference point, utility equals baseline utility from the realized outcome minus a penalty from failing, scaled by an individual loss-sensitivity parameter.[3] Raising the reference point increases the value of attaining a more demanding aspiration, but reduces the gain from exceeding it and increases the likelihood and size of the shortfall. Loss sensitivity parametrizes this trade-off by increasing the cost of failure without changing the reward from attaining a given aspiration. The optimal reference point balances these forces.

The optimal choice generates two reference-point regimes, depending on whether the chosen reference point can be missed. A reference point is *no-loss* when it is attained in every state and *aspirational* when some outcomes fall below it. For any finite lottery, when raising the reference point is not worthwhile even without a penalty for falling short, the agent remains in the no-loss regime regardless of loss sensitivity. When aspiring can be worthwhile, loss sensitivity determines which regime is optimal. Below a cutoff, the optimal reference point is aspirational; above it, it is no-loss. Which regime is optimal also depends on economic opportunities. In binary lotteries, a higher probability of the high outcome or greater payoff dispersion can make aspiration more attractive, so the same agent can choose a no-loss reference point in one environment and an aspirational one in another. At an interior aspirational optimum, both changes also raise the chosen reference point.

Which regime is chosen determines whether loss sensitivity affects the value of a lottery. In the no-loss regime, the value of the lottery is independent of loss sensitivity; in the aspirational regime, it decreases with loss sensitivity. Hence, two individuals who differ only in loss sensitivity can value one lottery the same and another differently when at least one of them chooses an aspirational reference point. Moreover, an individual may care strongly about falling short of an aspiration without this showing up in observed choice simply because she chooses a reference point that cannot be missed.

Unobserved reference points create an identification problem because observed choices depend

[3]For a fixed reference point, utility is concave above it, while its curvature below the reference point depends on the relative curvature of baseline utility and the shortfall penalty.

on both preferences and the reference point used to evaluate outcomes. In my model, however, the reference point is chosen optimally and cannot vary freely across lotteries. It must maximize expected utility under the same preferences in each choice problem. I use this optimality restriction together with a parametric specification to identify preferences. Since loss sensitivity matters only in the aspirational regime, while the optimal regime depends on the lottery, different lotteries can reveal different preference parameters. Under a specification in which the utility components share a curvature parameter, I show that a sufficiently low-dispersion lottery must be evaluated in the no-loss regime, independently of loss sensitivity. Its certainty equivalent therefore identifies curvature under this specification. Given this curvature, the model determines the no-loss certainty equivalent of a second lottery. If the observed certainty equivalent exceeds this benchmark, the agent must have chosen an aspirational reference point, and the second certainty equivalent identifies loss sensitivity.

So far, I have studied how the agent chooses a reference point for a given risky prospect, but the agent can also take actions that change the prospect itself. For a binary lottery, an action can change the payoffs while keeping their probabilities fixed, or change the probabilities while keeping the payoffs fixed. These two cases correspond to the two applications I study: in the portfolio choice application, the agent changes the payoffs across states, and in the effort provision application, effort changes the probability of the high outcome. In each application, the agent chooses the reference point and the action together. As a benchmark, I fix the reference point at its optimal value for the initial prospect and let the agent choose only the action. The difference isolates the effect of adjusting the reference point.

In the portfolio application, an investor allocates wealth across two equally likely states and chooses a wealth target.[4] The resulting allocation defines a binary lottery, where the investor chooses the two possible payoffs, trading off wealth in one state against wealth in the other. Full insurance corresponds to equal wealth in both states. When wealth in the high-payoff state becomes relatively cheaper, the investor shifts more wealth toward that state when her target can adjust than when it is held fixed. This is true in both regimes, although the target moves in opposite directions. In the no-loss regime, she lowers her target. Her surplus rises in both states, but the marginal value of wealth falls proportionally more in the low state, where the surplus is smaller. In the aspirational regime, she raises her target. Her surplus in the high state falls, and the marginal value of high-state wealth rises relative to low-state wealth. Either way, high-state wealth becomes relatively more valuable. I also show that loss sensitivity affects the portfolio only in the aspirational regime. At a stable interior optimum, a more loss-sensitive investor holds more low-state wealth and sets a lower target, which reinforces the shift toward full insurance.

In the effort application, I study how allowing aspirations to adjust changes the effect of loss

[4]Choosing wealth in each state is equivalent to buying Arrow–Debreu securities, one for each state. Choi et al. (2007) study this choice experimentally.

sensitivity on effort provision. A student chooses effort and an aspiration, and effort determines the probability of high performance. Holding an aspirational reference point fixed, greater loss sensitivity raises effort because avoiding failure becomes more valuable. When the reference point can adjust, however, a more loss-sensitive student can also lower her aspiration, which reduces the cost of failing. Reference-point adjustment can therefore work against the direct incentive to exert more effort. Under a parametric specification, the student chooses a no-loss reference point at low productivity, so loss sensitivity does not affect effort. Just above the productivity level at which aspirations become optimal, the chosen aspiration lies close to the low outcome. Missing it is therefore not very costly, so greater loss sensitivity provides little additional incentive to exert effort. Yet it still increases the marginal cost of raising the aspiration. The more loss-sensitive student then chooses a lower aspiration, which also reduces her effort because effort and aspirations are complements. At higher productivity, the chosen aspiration is higher, making failure more costly. The incentive to avoid failure eventually becomes stronger than the effect of lowering the aspiration, and greater loss sensitivity increases effor

In portfolio choice, reference-point adjustment amplifies the response to changes in relative prices. In effort choice, it can instead work against the direct effect of loss sensitivity and even reverse its direction. Holding aspirations fixed can therefore lead to incorrect predictions about the magnitude, or even the direction, of behavioral responses to changes in preferences and economic opportunities.

Related literature. This paper relates first to models of goal setting and aspirations. In Koch and Nafziger (2011) and Hsiaw (2013), goals help individuals overcome self-control problems arising from present bias. In Dalton et al. (2016), aspirations must be consistent with the outcomes produced by effort, and failure to internalize this feedback can generate low aspirations and low effort. Here, aspirations are optimally chosen and need not be attained, so individuals face a trade-off between aiming higher and the risk of falling short. Kim (2023) also studies joint goal and effort choice and shows that goals and effort are complements. In that model, goals provide direct value, whereas here the value of a higher aspiration depends on attaining it. I characterize aspiration choice for general risky prospects and show that, in the effort application, complementarity makes aspiration adjustment work against the direct incentive to increase effort, and can reverse its effect.

The paper is also related to Sarver (2018), who axiomatizes mixture-averse preferences with a representation in which individuals optimally select their risk attitudes. An earlier version, Sarver (2012), interprets this selection as optimal anticipation: anticipating a better outcome provides utility before uncertainty resolves, but increases disappointment when the outcome falls short.[5] In

[5]Sarver (2018) also studies implications for insurance demand and stock-market participation, and shows how the model can address the Rabin paradox. Brunnermeier and Parker (2005) study a related trade-off for optimal beliefs

my model, the benefit of choosing a higher aspiration depends on whether it is attained, whereas anticipatory utility in Sarver is received regardless of the realized outcome. My focus is on how preferences and economic opportunities determine the chosen aspiration, and how its adjustment affects actions and the identification of preferences.

In much of the reference-dependence literature, the reference point is taken as given. Diecidue and Van de Ven (2008) and Alaoui and Penta (2026), for example, study risk attitudes around exogenous aspiration levels. Other models specify how the reference point is determined by the choice environment or by the individual's behavior. In Masatlioglu and Ok (2014), the initial endowment provides the reference point, while in Kőszegi and Rabin (2006, 2007), it is determined by expectations consistent with choice. Kıbrıs et al. (2023) model reference-point formation through a conspicuity ranking, with the most conspicuous alternative in the choice set serving as the reference point. Empirically, Baillon et al. (2020) compare alternative reference-point rules and find that the status quo and a security level determined by the available lotteries are the most common. In my model, the reference point is instead chosen to maximize expected utility jointly with the lottery. Its level therefore responds to preferences and economic opportunities, and its adjustment affects the incentives to act.

Finally, an empirical literature shows that interventions aimed at raising aspirations can increase investment and economic activity (Bernard et al., 2026; Orkin et al., 2025), but can also have negative effects. McKenzie et al. (2022) find that an intervention raising financial aspirations led to lower borrowing and business investment, with most participants falling short of their goals.[6] These findings motivate studying not only how aspirations affect decisions, but also how the aspirations individuals choose depend on their economic opportunities and actions.

The remainder of the paper is organized as follows. Section 2 introduces the model and characterizes reference-point choice for finite lotteries. Section 3 studies binary lotteries and derives implications for risky choice. Section 4 addresses identification when reference points are unobserved. Section 5 studies portfolio choice and effort provision under endogenous aspirations. Section 6 concludes.

rather than optimal reference points.

[6]Related evidence on aspirations, goals, and economic outcomes includes Beaman et al. (2012); Carlana et al. (2022); Bernard et al. (2019); Riley (2024); Goerg and Kube (2012). Aspirations also respond to labor-market conditions (Molina and Usui, 2023). For reviews, see La Ferrara (2019); Genicot and Ray (2020).

2 Model

2.1 Reference-dependent utility

Research in psychology suggests that satisfaction depends both on the difficulty of the goal attained and on the performance relative to that goal. Meyer et al. (1988) find that the people expect more satisfaction from attaining a harder goal, even though it is less likely to be reached. At the same time, satisfaction also depends on how far performance exceeds or falls short of the goal (Locke and Latham, 2002) and for a given performance, more demanding goals are associated with lower satisfaction (Meyer et al., 1988; Mento et al., 1992). Heath et al. (1999) shows that goals also act as reference points and have both loss aversion and diminishing sensitivity around them. These findings suggest a trade-off in goal setting: a higher goal can be more rewarding when attained, but leaves less surplus above it and increases the likelihood and cost of falling short. The idea that the choice of a goal balances the attraction of success against the cost of failure goes back to Lewin et al. (1944).

To study how individuals choose such goals or aspirations, and how these choices affect behavior under uncertainty, I develop a model that characterizes this trade-off. A decision maker chooses a reference point r before uncertainty resolves and then receives a realized outcome x. The reference point is an outcome threshold: outcomes above r are experienced as successes and outcomes below r as failures.[7] Outcomes and reference points are measured in the same units, with $x, r \in \mathbb{R}_+$. The reference point is chosen from a feasible set of reference points induced by the menu, which I define in Section 2.2. For now, fix a pair (x, r). The utility depends on whether the realized outcome meets the reference point:

$$U(x,r) = \begin{cases} u(r) + g(x-r), & \text{if } x \geq r, \\ u(x) - \lambda \ell(r-x), & \text{if } x < r, \end{cases} \qquad \lambda \geq 0. \tag{1}$$

Assumption 1 (Primitive utility functions)**.** *The functions $u, g, \ell : \mathbb{R}_+ \to \mathbb{R}$ are twice continuously differentiable, strictly increasing, with*

$$u'', g'' < 0, \ell'' \leq 0 \qquad g(0) = \ell(0) = 0, \qquad u'(0),\ g'(0),\ \ell'(0) < \infty.$$

When the outcome reaches the reference point ($x \geq r$), utility depends on both satisfaction from attaining the aspiration $u(r)$, and the utility from exceeding it, $g(x-r)$. When the outcome falls below the reference point ($x < r$), utility equals the baseline utility from the realized outcome minus a penalty from failing, scaled by the loss-sensitivity parameter λ. Raising the reference point has

[7] Aspirations or goals chosen by the agent are one possible interpretation of such a threshold (Heath et al., 1999).

an ambiguous effect when it is attained, since $u'(r) - g'(x - r)$ need not be positive, but when it is missed, a higher reference point always increases the shortfall penalty. A higher reference point is worth choosing only if the value of attaining a more demanding aspiration outweighs the smaller gain from exceeding it and the higher cost of falling short. Loss sensitivity parametrizes this trade-off by increasing the cost of failure without changing the reward from attaining a given aspiration. Figure 1 illustrates the utility function around a chosen reference point.

The asymmetric specification is what creates the trade-off in choosing the reference point. Consider a conventional reference-dependent specification of the form $u(x) + v(x - r)$, where v is increasing. The utility is then decreasing in r for every realized outcome, so an agent who can choose the reference point would always select the lowest feasible one. Allowing the utility to depend on the level of the attained reference point introduces a benefit from choosing a more demanding aspiration. Suppose instead that utility in the loss domain were anchored at $u(r)$, as in $u(r) - \lambda\ell(r - x)$. A higher reference point could then raise utility even when the aspiration is missed, since the increase in $u(r)$ could exceed the additional shortfall penalty. Anchoring utility in the loss domain at $u(x)$ rules this out and ensures that a higher reference point can be beneficial only when it is attained.

The function g measures the gain from exceeding the reference point, and ℓ the cost of falling short. Both are concave which implies diminishing sensitivity: an additional unit above or below r matters less the further the outcome is from it (weakly for ℓ). Because u appears in both domains and $g(0) = \ell(0) = 0$, $U(x, r)$ is continuous at $x = r$.

Concavity of ℓ does not imply that utility is concave below the reference point. In the loss domain, utility is equal to the strictly concave baseline utility $u(x)$ minus a concave shortfall penalty, so

$$U_{xx}(x, r) = u''(x) - \lambda\ell''(r - x).$$

Since $\ell'' \leq 0$, the second term is nonnegative and can dominate the concavity of u. Hence, utility can be locally convex below the reference point even though the shortfall function is concave. When ℓ is linear, the loss domain is strictly concave.

2.2 Choice problem and feasible reference points

Timing and primitives. Let $\mathcal{X} = \{x_1, \ldots, x_K\} \subset \mathbb{R}_+$ be a finite set of outcomes, and let $\Delta(\mathcal{X})$ denote the set of lotteries over $\mathcal{X}$. A menu is a nonempty finite set $M \subseteq \Delta(\mathcal{X})$. After observing M, the decision maker jointly chooses a lottery $p \in M$ and a reference point r. Nature then draws an outcome $x \in \mathcal{X}$ with probability $p(x)$, and the decision maker gets utility $U(x, r)$.[8] For $p \in \Delta(\mathcal{X})$, define $\text{supp}(p) := \{x \in \mathcal{X} : p(x) > 0\}$, $x_\ell(p) := \min \text{supp}(p)$, $x_h(p) := \max \text{supp}(p)$. For a menu

[8]The agent chooses the reference point before uncertainty resolves and does not revise it after seeing the outcome. It can differ across decisions.

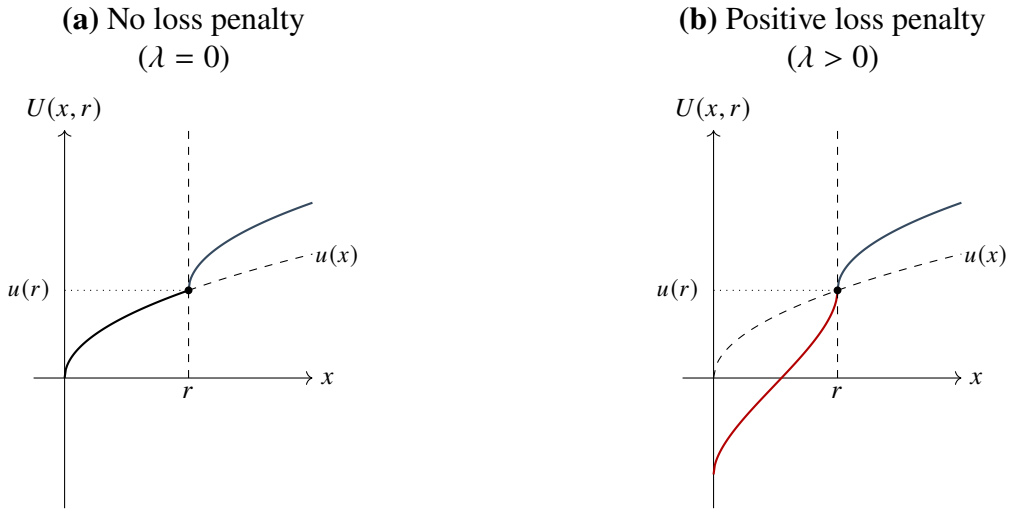


Figure 1: Reference-dependent utility for a given reference point r. In panel (a), $\lambda = 0$, so utility is $u(x)$ below r and $u(r) + g(x - r)$ above it. In panel (b), $\lambda > 0$, and utility below r falls by the shortfall penalty $\lambda\ell(r - x)$.

M, define its lowest and highest outcomes by

$$\underline{x}(M) := \min_{p \in M} x_\ell(p), \qquad \bar{x}(M) := \max_{p \in M} x_h(p).$$

Feasible reference points. I restrict reference points to the range of outcomes available in the menu. Any threshold within this range can serve as a reference point, even if it is not itself a possible realization of a lottery. The lower bound rules out reference points below every available outcome, which could otherwise be optimal because the gain in surplus utility outweighs the reduction in utility from attaining a lower reference point. The upper bound is without loss of generality. A reference point above the highest outcome in the menu cannot improve utility relative to that outcome. It also ensures that the feasible set of reference points is compact.[9]

Assumption 2 (Menu-range feasibility)**.** *For every finite menu M, the feasible reference-point set is common across alternatives and equals*

$$\mathcal{R}(M) := [\underline{x}(M), \bar{x}(M)].$$

The menu fixes which reference points are available, and the chosen lottery determines how likely a given reference point is to be attained. Since $\underline{x}(M) \leq x_\ell(p)$ for every $p \in M$, every lottery in the menu can be paired with a reference point at or below its lowest outcome, which it attains

[9]Restricting aspirations to available opportunities is consistent with the idea that aspirations adapt to the decision environment (Selten, 1998).

with certainty. All lotteries share the same feasible set, but their optimal reference points can differ. Because this set depends on the menu, adding or removing alternatives can change the value of a lottery that is itself unchanged.

Indirect values and choice over lotteries. After observing a menu M, the decision maker jointly chooses a lottery and a feasible reference point:

$$\max_{p \in M,\, r \in \mathcal{R}(M)} \mathbb{E}_p[U(X, r)].$$

For each $p \in M$, define its indirect value by

$$V_M(p; \lambda) := \max_{r \in \mathcal{R}(M)} \mathbb{E}_p[U(X, r)],$$

and let

$$r^*_M(p, \lambda) \in \arg\max_{r \in \mathcal{R}(M)} \mathbb{E}_p[U(X, r)]$$

denote a selected optimal reference point. This formulation does not mean that the agent chooses p first and r second. The value $V_M(p; \lambda)$ is the expected utility of p under its best feasible reference point, so the joint problem is equivalent to

$$p^*_M(\lambda) \in \arg\max_{p \in M} V_M(p; \lambda).$$

The pair $\big(p^*_M(\lambda), r^*_M\big(p^*_M(\lambda), \lambda\big)\big)$ solves the joint choice problem. A solution exists under Assumptions 1 and 2. For each $p \in M$, $\mathbb{E}_p[U(X, r)]$ is continuous in r and $\mathcal{R}(M)$ is compact, so an optimal reference point exists. Because M is finite, an optimal lottery exists as well.

For a fixed menu M, define the induced preference over lotteries by

$$p \succeq_M q \quad \Longleftrightarrow \quad V_M(p; \lambda) \geq V_M(q; \lambda).$$

Because V_M is computed over $\mathcal{R}(M)$, preferences over lotteries can depend on the menu. They still respect first-order stochastic dominance. By Assumption 1, $U(x, r)$ is strictly increasing in x for every fixed r. Hence, if p first-order stochastically dominates q, then

$$\mathbb{E}_p[U(X, r)] \geq \mathbb{E}_q[U(X, r)] \qquad \text{for every } r \in \mathcal{R}(M).$$

Since Assumption 2 gives every lottery the same feasible set, maximizing both sides over $\mathcal{R}(M)$ gives $V_M(p; \lambda) \geq V_M(q; \lambda)$, that is, $p \succeq_M q$.

2.3 No-loss and aspirational regimes

Consider the choice of reference point for a given lottery p. Fix a menu M and a nondegenerate lottery $p \in M$, with $x_\ell(p) < x_h(p)$. Reference points above $x_h(p)$ can be ignored.[10] Define

$$\mathcal{R}_M^{NL}(p) := \mathcal{R}(M) \cap (-\infty, x_\ell(p)], \qquad \mathcal{R}_M^{A}(p) := \mathcal{R}(M) \cap (x_\ell(p), x_h(p)].$$

Reference points in $\mathcal{R}_M^{NL}(p)$ are *no-loss*: every realization of p weakly exceeds the reference point, so the loss domain is never reached. Reference points in $\mathcal{R}_M^{A}(p)$ are *aspirational*: some realizations attain the reference point and others fail to reach it.

These labels depend on the lottery, not on the level of the reference point: they record whether the reference point can be missed under that lottery. An aspirational reference point for one lottery can lie below a no-loss reference point for another. Within a given lottery, every aspirational reference point lies above every no-loss one. The best no-loss reference point need not be the lowest feasible one. Raising r within the no-loss range increases $u(r)$ but lowers the gain from exceeding it in every outcome.

Regime-specific values. For any $r \in \mathcal{R}_M^{NL}(p)$, all outcomes lie in the gain domain, and

$$\mathbb{E}_p[U(X, r)] = u(r) + \mathbb{E}_p[g(X - r)]. \tag{2}$$

Define the no-loss indirect value by

$$V_M^{NL}(p) := \max_{r \in \mathcal{R}_M^{NL}(p)} \left\{ u(r) + \mathbb{E}_p[g(X - r)] \right\}.$$

Because the loss term is never realized, $V_M^{NL}(p)$ is independent of λ.

Lemma 1 (Unique no-loss reference point)**.** *Suppose Assumptions 1 and 2 hold. For every menu M and every lottery $p \in M$, the objective $u(r) + \mathbb{E}_p[g(X - r)]$ is strictly concave in r on $\mathcal{R}_M^{NL}(p)$. Hence, the no-loss maximizer exists and is unique. Denote it by $r_M^{NL}(p)$.*

For any $r \in \mathcal{R}_M^{A}(p)$, the expected utility can be written as

$$\mathbb{E}_p[U(X, r)] = G_p(r) - \lambda L_p(r),$$

[10] For $\lambda > 0$, expected utility is strictly decreasing in r for any $r > x_h(p)$. If $\lambda = 0$, all $r > x_h(p)$ give the same expected utility as $r = x_h(p)$. So it is without loss to restrict attention to $r \leq x_h(p)$.

where

$$G_p(r) := \mathbb{E}_p\big[\big(u(r) + g(X - r)\big)\mathbf{1}\{X \geq r\} + u(X)\mathbf{1}\{X < r\}\big],$$
$$L_p(r) := \mathbb{E}_p\big[\ell(r - X)\mathbf{1}\{X < r\}\big].$$

The term $G_p(r)$ is the expected utility from r, excluding the penalty from failing. The term $L_p(r)$ is the expected penalty from failing and is strictly positive for every $r \in \mathcal{R}_M^A(p)$.

An aspirational reference point r is preferred to the best no-loss reference point if and only if

$$G_p(r) - V_M^{NL}(p) > \lambda L_p(r).$$

The left-hand side captures the value gained from moving the reference point above the no-loss reference-point, before subtracting the penalty from failing. Aspirations are therefore optimal only if this gain is positive and large enough to surpass the expected penalty from failing. Define the aspirational value by

$$V_M^A(p, \lambda) := \sup_{r \in \mathcal{R}_M^A(p)} \left\{G_p(r) - \lambda L_p(r)\right\}.$$

I use the supremum because $\mathcal{R}_M^A(p)$ is open at $x_\ell(p)$. When the loss sensitivity increases, the aspirational indirect value declines. So for sufficiently high loss sensitivity, the aspirational indirect value tends to the value obtained at the no-loss boundary $r = x_\ell(p)$. The indirect value of lottery p can then be written as

$$V_M(p; \lambda) = \max\left\{V_M^{NL}(p), V_M^A(p, \lambda)\right\}.$$

Theorem 1 (Cutoff in loss sensitivity)**.** *Let Assumptions 1 and 2 hold. Fix a menu M and a nondegenerate lottery $p \in M$. Then $V_M^A(p, \lambda)$ is finite, convex, continuous, and weakly decreasing in λ. Define*

$$\bar{\lambda}_M(p) := \inf\left\{\lambda \geq 0 : V_M^A(p, \lambda) \leq V_M^{NL}(p)\right\} < \infty.$$

Then:

1. *if $\lambda < \bar{\lambda}_M(p)$, every optimal reference point is aspirational;*
2. *if $\lambda > \bar{\lambda}_M(p)$, the unique optimal reference point is no-loss and equals $r_M^{NL}(p)$;*
3. *if $\lambda = \bar{\lambda}_M(p)$, the no-loss reference point $r_M^{NL}(p)$ is optimal, and aspirational reference points may also tie.*

The theorem shows that each lottery has a cutoff in loss sensitivity. Below the cutoff, the agent chooses an aspirational reference point; above it, the agent chooses a no-loss reference point. Once the no-loss reference point is chosen, the value of the lottery is independent of loss sensitivity

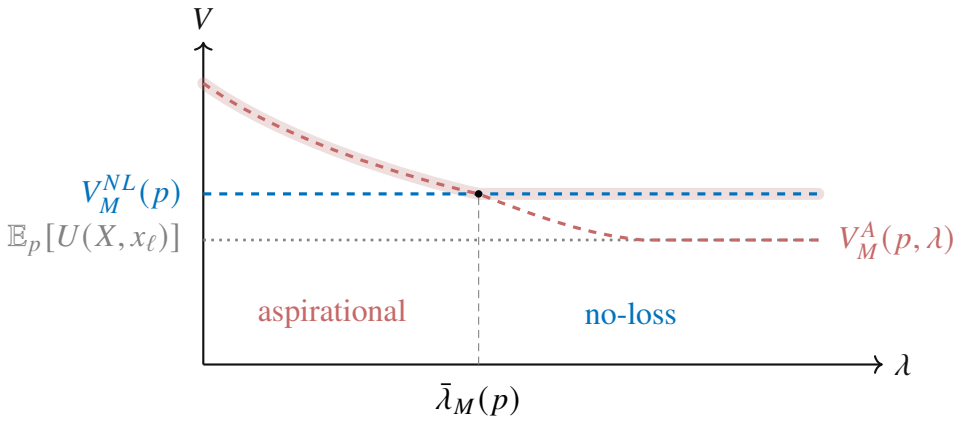


Figure 2: Aspirational and no-loss regimes. The aspirational value $V_M^A(p, \lambda)$ is decreasing and convex and equals $\mathbb{E}_p[U(X, x_\ell)]$ for sufficiently large λ. The no-loss value $V_M^{NL}(p)$ is constant and weakly exceeds $\mathbb{E}_p[U(X, x_\ell)]$. If x_ℓ is the no-loss maximizer, then $V_M^{NL}(p) = \mathbb{E}_p[U(X, x_\ell)]$. The optimal regime switches at $\bar{\lambda}_M(p)$. The value of the lottery is the upper envelope of the two branches, shown in pink.

because no outcomes fall in the loss domain. The no-loss value may be optimal even at $\lambda = 0$, which would imply $\bar{\lambda}_M(p) = 0$.

To see why this is the case, fix an aspirational reference point r. The expected utility is an affine function in λ, with a strictly negative slope $-L_p(r)$. The aspirational envelope $V_M^A(p, \lambda)$ is therefore weakly decreasing and convex. On the other hand, the no-loss value $V_M^{NL}(p)$ is constant in λ. As loss sensitivity increases, aspirations become less attractive relative to the no-loss reference point. Once the no-loss reference point becomes optimal, an additional increase in λ cannot make aspirations optimal again.

The cutoff in Theorem 1 is finite. Lemma A.2 in the appendix shows that, for sufficiently high λ, no aspirational reference point gives a higher value than the reference point $x_\ell(p)$. The intuition is that raising the reference point above $x_\ell(p)$ imposes a strictly positive penalty whenever the outcome is $x_\ell(p)$, which occurs with positive probability. As loss sensitivity increases, this penalty eventually outweighs any benefit from choosing a higher reference point. Since $x_\ell(p)$ is always feasible in the no-loss regime and the optimal no-loss reference point performs at least as well as $x_\ell(p)$, the no-loss regime must be optimal for all sufficiently high λ. The no-loss value is independent of λ, while the aspirational value is continuous in λ. Since the no-loss regime is optimal for every λ above the cutoff, continuity implies that its value cannot fall below the aspirational value at the cutoff itself, so $r_M^{NL}(p)$ is optimal at $\bar{\lambda}_M(p)$, while an aspirational reference point can also tie.

Proposition 1 (Menu minimum and reference-point choice)**.** *Let $M \subseteq M'$ and $p \in M$. For every $\lambda \geq 0$:*

1. *If $\underline{x}(M') = \underline{x}(M)$, then $V_{M'}(p; \lambda) = V_M(p; \lambda)$.*

2. *If* $\underline{x}(M') < \underline{x}(M)$, *then* $V_{M'}(p;\lambda) \geq V_M(p;\lambda)$ *and* $\bar{\lambda}_{M'}(p) \leq \bar{\lambda}_M(p)$.

Lowering the menu minimum expands the set of feasible no-loss reference points without changing the value attainable with an aspirational reference point. It can therefore increase the value of a lottery and move its optimal evaluation from aspirational to no-loss, but not in the opposite direction. However, raising the menu maximum cannot increase the lottery's value, since reference points above the highest outcome of the lottery's support cannot improve upon choosing that outcome as the reference point.

3 Binary Lotteries: Endogenous Activation of Loss Sensitivity

I now specialize the analysis to binary lotteries. With two outcomes, every aspirational reference point leaves the high outcome as a success and the low outcome as a failure, so changing the reference point does not change the probability of success. The cutoff in loss sensitivity then has a closed form, which shows how it depends on the features of the lottery.

Consider a binary lottery $p = (x_h, \pi; x_\ell, 1-\pi)$, with $0 < x_\ell < x_h$ and probability of success $\pi \in (0,1)$. The agent chooses between p and a sure payoff $y \in [x_\ell, x_h]$. Let $M(y) = \{p, \delta_y\}$. By Assumption 2, the feasible set of reference points is $[x_\ell, x_h]$. Since varying y does not change this set, the value of the risky lottery is independent of y. For simplicity, I write it as $V(p,\lambda)$.

3.1 Binary reference-point choice

For any reference point $r \in [x_\ell, x_h]$, the expected utility of the risky lottery is

$$\mathbb{E}_p[U(X,r)] = \pi\big[u(r) + g(x_h - r)\big] + (1-\pi)\big[u(x_\ell) - \lambda\ell(r - x_\ell)\big]. \tag{3}$$

The reference point $r = x_\ell$ is no-loss, while every $r \in (x_\ell, x_h]$ is aspirational. At $r = x_\ell$, $E_p[U(X,r)]$ is independent of λ.

For any aspirational $r > x_\ell$, the difference in expected utility at r relative to the no-loss reference point is

$$\mathbb{E}_p[U(X,r)] - \mathbb{E}_p[U(X,x_\ell)] = \pi \underbrace{\Big[u(r) + g(x_h - r) - u(x_\ell) - g(x_h - x_\ell)\Big]}_{\text{potential aspiration gain}} - (1-\pi)\underbrace{\lambda\ell(r - x_\ell)}_{\text{aspiration cost}}.$$

The first term is the potential gain from adopting the higher reference point when the high outcome

realizes. The second is the expected cost of missing it when the low outcome realizes. An aspirational reference point is then preferred to a no-loss one iff

$$\lambda < \frac{\pi}{1-\pi} \frac{u(r) + g(x_h - r) - u(x_\ell) - g(x_h - x_\ell)}{\ell(r - x_\ell)}.$$

The agent chooses an aspirational reference point whenever at least one $r > x_\ell$ satisfies this inequality. Define the following aspiration index

$$A^*(p) := \sup_{r \in (x_\ell, x_h]} \frac{u(r) + g(x_h - r) - u(x_\ell) - g(x_h - x_\ell)}{\ell(r - x_\ell)}.$$

The index $A^*(p)$ is the highest gain that can be obtained from an aspiration with respect to a no-loss reference point per unit of shortfall cost under the lottery p before accounting for probabilities. It is nonpositive exactly when, at $r = x_\ell$, the marginal utility of surplus is at least the marginal utility of raising the reference point, $g'(x_h - x_\ell) \geq u'(x_\ell)$. When $A^*(p) > 0$, the loss-sensitivity cutoff is

$$\bar{\lambda}(p) := \frac{\pi}{1-\pi} A^*(p).$$

Proposition 2 (Binary cutoff with the global aspiration index)**.** *Maintain Assumptions 1 and 2. Fix a binary lottery $p = (x_h, \pi; x_\ell, 1-\pi)$ with $0 < x_\ell < x_h$ and $\pi \in (0,1)$. Let $A^*(p)$ be defined as above.*

1. *If $A^*(p) \leq 0$, the no-loss reference point $r = x_\ell$ is optimal for every $\lambda \geq 0$. If $A^*(p) = 0$, an aspirational reference point can also tie when $\lambda = 0$.*

2. *If $A^*(p) > 0$, then there exists $\bar{\lambda}(p) > 0$ such that*

$$r^*(p, \lambda) \in \begin{cases} (x_\ell, x_h], & \text{if } \lambda < \bar{\lambda}(p), \\ \{x_\ell\}, & \text{if } \lambda > \bar{\lambda}(p). \end{cases}$$

At $\lambda = \bar{\lambda}(p)$, x_ℓ is optimal and an aspirational reference point can also tie.

Proposition 2 gives a direct expression for the loss-sensitivity threshold from Theorem 1 above which the agent stops aspiring. The cutoff depends on the success odds ratio and on the aspiration index $A^*(p)$. If $A^*(p) \leq 0$, no aspirational reference point gives a higher value than x_ℓ even when $\lambda = 0$, so the agent chooses $r = x_\ell$ for every λ. If $A^*(p) > 0$, an aspirational reference point is optimal below $\bar{\lambda}(p)$ and $r = x_\ell$ is optimal above it.

For the comparative statics below, I use a shifted-power specification, equivalent to the shifted CRRA family in Apesteguia et al. (2024). I also impose a common-shift and a curvature ordering $\alpha_\ell \leq \alpha_u$ which guarantees uniqueness of the aspirational reference point.

Assumption 3 (Common-shift shifted-power specification)**.** *Let*

$$v_{\alpha,\sigma}(z) = \begin{cases} \dfrac{(z+\sigma)^{\alpha} - \sigma^{\alpha}}{\alpha}, & \alpha \neq 0, \\ \log\left(\dfrac{z+\sigma}{\sigma}\right), & \alpha = 0. \end{cases}$$

For some $\sigma > 0$, $\alpha_u, \alpha_g, \alpha_\ell < 1$ and $\alpha_\ell \leq \alpha_u$ the primitives satisfy $(u, g, \ell) = \left(v_{\alpha_u,\sigma}, v_{\alpha_g,\sigma}, v_{\alpha_\ell,\sigma}\right)$.

Lemma 2 (Uniqueness of the aspirational reference point)**.** *Maintain Assumptions 1, 2, and 3. Fix a binary lottery $p = (x_h, \pi; x_\ell, 1-\pi)$ with $0 < x_\ell < x_h$ and $\pi \in (0,1)$. Then, for every $\lambda \geq 0$ the expected utility $\mathbb{E}_p[U(X, r)]$ has at most one local maximum on $(x_\ell, x_h]$.*

The curvature ordering from Assumption 3 implies that the second-order condition of the reference point objective can change sign at most once. This rules out multiple aspirational local maxima.

3.2 Certainty equivalents and risky choice

Risky choice compares the lottery with a sure payoff, which is also evaluated at its optimal reference point. Its indirect value is

$$S(y) := V_{M(y)}(\delta_y; \lambda) = \max_{r \in [x_\ell, y]} \{u(r) + g(y - r)\}. \tag{4}$$

The value $S(y)$ is independent of λ, continuous, and strictly increasing in y. Moreover, p is strictly preferred to δ_{x_ℓ}, while δ_{x_h} is strictly preferred to p. Hence, there is a unique certainty equivalent $\bar{y}(p; \lambda)$ satisfying $S(\bar{y}(p; \lambda)) = V(p; \lambda)$,

$$p \succeq_{M(y)} \delta_y \iff y \leq \bar{y}(p; \lambda).$$

The no-loss value of the lottery is $V^{NL}(p) = u(x_\ell) + \pi g(x_h - x_\ell)$. Let $\bar{y}^{NL}(p)$ denote the corresponding no-loss certainty equivalent, defined by

$$S\left(\bar{y}^{NL}(p)\right) = V^{NL}(p).$$

Proposition 3 (Certainty-equivalent plateau)**.** *Fix a nondegenerate binary lottery p.*

1. *If $A^*(p) \leq 0$, then $\bar{y}(p; \lambda) = \bar{y}^{NL}(p)$ for every $\lambda \geq 0$.*
2. *If $A^*(p) > 0$, then $\bar{y}(p; \lambda)$ is strictly decreasing for $\lambda < \bar{\lambda}(p)$ and $\bar{y}(p; \lambda) = \bar{y}^{NL}(p)$ for $\lambda \geq \bar{\lambda}(p)$.*

When the reference point is aspirational, greater loss sensitivity lowers the value of the risky lottery and therefore lowers its certainty equivalent. Once $r = x_\ell$ is chosen, the valuation of the lottery no longer depends on λ, so the certainty equivalent is constant.

Corollary 1 (Pooling and separation)**.** *Fix two types $0 < \lambda_1 < \lambda_2$ who share the same primitives (u, g, ℓ).*

1. *If $A^*(p) \leq 0$, or if $A^*(p) > 0$ and $\bar{\lambda}(p) \leq \lambda_1$, then $\bar{y}(p; \lambda_1) = \bar{y}(p; \lambda_2) = \bar{y}^{NL}(p)$.*
2. *If $A^*(p) > 0$ and $\lambda_1 < \bar{\lambda}(p)$, then $\bar{y}(p; \lambda_1) > \bar{y}(p; \lambda_2)$.*

The same two types can therefore have the same certainty equivalent for one lottery and different certainty equivalents for another one. This happens because loss sensitivity affects the certainty equivalent only when an aspirational reference point is chosen, and in some environments, agents with similar risk attitudes can choose the same no-loss reference point, even if their sensitivity to failure is different.

Corollary 2 (Sorting into risky choice)**.** *Fix $0 < \lambda_1 < \lambda_2$. The two types make different choices between p and δ_y if and only if*

$$\bar{y}(p; \lambda_2) < y < \bar{y}(p; \lambda_1).$$

So two types who differ only in λ choose differently only if the sure payoff lies between their certainty equivalents, which requires at least one of them to aspire.

Comparison with expectations-based reference points. The certainty-equivalent plateau does not arise in the canonical expectations-based model of Kőszegi and Rabin (2007). Under choice-acclimating personal equilibrium with piecewise-linear gain–loss utility, the value of a nondegenerate binary lottery strictly decreases with loss aversion. The difference is not only due to the utility specification. With my utility function, evaluating each lottery against its own outcome distribution, rather than against an optimally chosen aspiration, also gives a value that strictly decreases with loss sensitivity. In both benchmarks, the reference point is drawn from the lottery itself. With probability $\pi(1 - \pi)$, the outcome is low while the reference point is high so it falls in the loss domain. Here, the agent instead chooses a scalar reference point and can set it at x_ℓ, which is attained in every state. When loss sensitivity is above the cutoff $\bar{\lambda}(p)$, this choice is optimal and the certainty equivalent does not depend on loss sensitivity. Appendix A.4 provides both calculations.

3.3 Activation margins

Success probability and payoff dispersion both affect whether the agent chooses an aspirational reference point, and either can move the same agent from one regime to the other.

3.3.1 Success probability

Fix (x_ℓ, x_h) and let

$$p_\pi = (x_h, \pi; x_\ell, 1-\pi).$$

Since the aspiration index depends only on the support, write it as $A^*(x_\ell, x_h)$ and suppose $A^*(x_\ell, x_h) > 0$. For a fixed type λ, the cutoff can be written as

$$\pi^A(\lambda) := \frac{\lambda}{\lambda + A^*(x_\ell, x_h)}.$$

The agent chooses an aspirational reference point if and only if $\pi > \pi^A(\lambda)$. For two types $0 < \lambda_1 < \lambda_2$, $\pi^A(\lambda_1) < \pi^A(\lambda_2)$, so the less loss-sensitive type starts aspiring first. Write $\bar{y}(\pi; \lambda) := \bar{y}(p_\pi; \lambda)$ and define

$$\Delta_y(\pi; \lambda_1, \lambda_2) := \bar{y}(\pi; \lambda_1) - \bar{y}(\pi; \lambda_2).$$

Proposition 4 (Success probability and separation)**.** *Fix a support (x_ℓ, x_h) with $A^*(x_\ell, x_h) > 0$ and two types $0 < \lambda_1 < \lambda_2$.*

1. *If $\pi \le \pi^A(\lambda_1)$, both types choose $r = x_\ell$ and $\Delta_y(\pi; \lambda_1, \lambda_2) = 0$.*
2. *If $\pi > \pi^A(\lambda_1)$, then $\Delta_y(\pi; \lambda_1, \lambda_2) > 0$.*
3. *As $\pi \to 1$, $\Delta_y(\pi; \lambda_1, \lambda_2) \to 0$.*

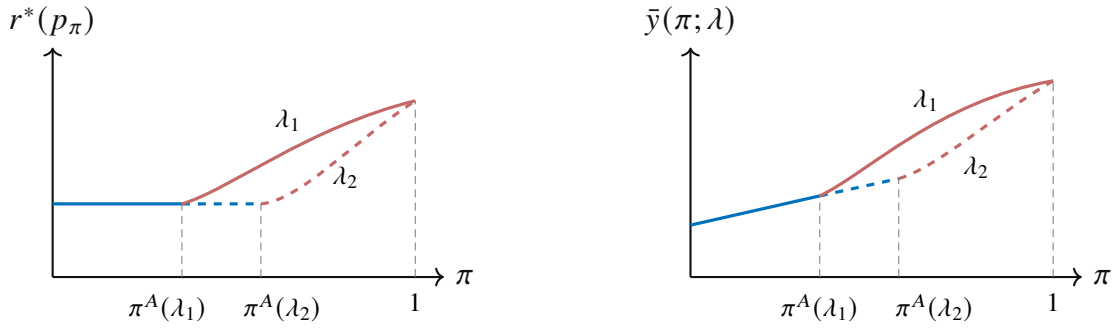
Figure 3 illustrates these changes in the reference point and the certainty equivalent. At low success probability, the two types choose the same no-loss reference point, and have the same certainty equivalent. As the probability of success increases, the type with a lower cost of penalty from failing will start aspiring first and their certainty equivalents differ. As the probability of success goes to one, the gap goes back to zero as less weight is placed on the loss domain.

3.3.2 Dispersion

Next, vary dispersion while keeping expected value and the probability of success fixed. Fix $\mu > 0$ and $\pi \in (0, 1)$, and write

$$x_h(d) = \mu + (1-\pi)d, \qquad x_\ell(d) = \mu - \pi d, \qquad d \in [0, \mu/\pi).$$

Increasing d raises the high outcome and lowers the low outcome while keeping the expected payoff fixed. Throughout this subsection, I suppress the fixed parameters (μ, π) from the notation. Let



(a) Optimal reference point *(b) Certainty equivalent*

Figure 3: Success probability and activation. Since $\pi^A(\lambda_1) < \pi^A(\lambda_2)$, the λ_1 type becomes aspirational first. At low success probabilities, both types choose $r = x_\ell$ and have the same certainty equivalent. Once the λ_1 type starts aspiring, their certainty equivalents differ. As $\pi \to 1$, the gap goes back to zero. Blue segments indicate a no-loss reference point and red segments an aspirational one; solid lines show λ_1 and dashed lines λ_2.

$p(d) = (x_h(d), \pi; x_\ell(d), 1-\pi)$. Its aspiration index is

$$A^*(d) = \sup_{t\in(0,d]} \frac{u(\mu - \pi d + t) + g(d-t) - u(\mu - \pi d) - g(d)}{\ell(t)}.$$

When $A^*(d) > 0$, the loss-sensitivity cutoff is

$$\bar{\lambda}(d) = \frac{\pi}{1-\pi} A^*(d).$$

Proposition 5 (Dispersion and aspiration)**.** *Maintain Assumptions 1 and 2.*

The aspiration index $A^*(d)$, and therefore $\bar{\lambda}(d)$ whenever $A^*(d) > 0$, is strictly increasing in d. For any $\lambda > 0$, define

$$\bar{d}(\lambda) := \inf\{d : \bar{\lambda}(d) > \lambda\}.$$

For $d < \bar{d}(\lambda)$, the agent chooses $r = x_\ell(d)$. For $d > \bar{d}(\lambda)$, every optimal reference point is aspirational. At $d = \bar{d}(\lambda)$, $x_\ell(d)$ is optimal and an aspirational reference point can also tie.

If $g = u$, all types choose $r = x_\ell(d)$ whenever

$$d \le d^G(\mu, \pi) := \frac{\mu}{1+\pi}.$$

Greater dispersion therefore makes aspiration attractive for a larger set of loss-sensitivity types. When $g = u$, $d^G(\mu, \pi)$ gives a common low-dispersion region in which all types choose $r = x_\ell(d)$. Above that region, the dispersion at which the agent starts aspiring depends on λ, with lower-λ types

aspiring first. For two types $0 < \lambda_1 < \lambda_2$, write $\bar{y}(d;\lambda) := \bar{y}(p(d);\lambda)$ and define

$$\Delta_y(d;\lambda_1,\lambda_2) := \bar{y}(d;\lambda_1) - \bar{y}(d;\lambda_2).$$

Corollary 3 (Dispersion and separation)**.** *Fix* $0 < \lambda_1 < \lambda_2$.

1. *If* $d \le \bar{d}(\lambda_1)$*, then* $\Delta_y(d;\lambda_1,\lambda_2) = 0$.
2. *If* $d > \bar{d}(\lambda_1)$*, then* $\Delta_y(d;\lambda_1,\lambda_2) > 0$.

The two types therefore have the same certainty equivalent at low dispersion, and their certainty equivalents differ once dispersion is high enough for the less loss-sensitive type to start aspiring.

Lemma 3 (Dispersion and the chosen reference point)**.** *Fix a type* $\lambda > 0$*, and let* $r_p^*(d,\lambda)$ *be the reference point used to evaluate* $p(d)$*. On the no-loss branch,* $r_p^*(d,\lambda) = x_\ell(d) = \mu - \pi d$*, so the chosen reference point falls with dispersion. On any differentiable interior aspirational branch,*

$$\frac{d}{dd} r_p^*(d,\lambda) > 0.$$

Dispersion therefore moves the reference point in opposite directions across the two regions. When $r = x_\ell(d)$, it falls with the low outcome. Once the agent aspires, greater dispersion instead raises the chosen reference point.

4 Identification

Preferences can be identified even when reference points are not observed. Observed choices depend on both preferences and the reference point used to evaluate outcomes. However, in my model, the reference point is chosen optimally and cannot vary freely across choice problems. For any lottery, it must maximize expected utility given the agent's preferences. This restricts how reference points can change across lotteries and allows preferences to be identified without observing them.

When the reference point is no-loss, it is attained in every state, so loss sensitivity does not affect valuation. When it is aspirational, some outcomes fall below it, and loss sensitivity matters. This allows different lotteries to identify different preference parameters.

I show this result under a specification in which the three utility components are represented by the same shifted-power function from Assumption 3, $u = g = \ell = v_{\alpha,\sigma}$, with $\alpha < 1$ and known $\sigma > 0$. Since σ is known, I write $v_\alpha := v_{\alpha,\sigma}$. The parameters to identify are curvature α and loss sensitivity λ.

A sufficiently low-dispersion lottery is evaluated in the no-loss regime for every α and λ, so its certainty equivalent pins down curvature alone. Once curvature is known, the model determines the

no-loss certainty equivalent of a second lottery. If the observed certainty equivalent lies strictly above this benchmark, the agent must have chosen an aspirational reference point. Since the certainty equivalent is strictly decreasing in loss sensitivity in this regime, the observation uniquely identifies λ.

The elicitation can be designed in a way that keeps the feasible set of reference points fixed. Let the agent chooses between a lottery p_n and a sure payoff y, with y varied over $[x_{\ell n}, x_{hn}]$. Because y never leaves this interval, every menu has the same lowest and highest outcomes, $x_{\ell n}$ and x_{hn}, and the feasible reference-point set is

$$\mathcal{R}(M_n(y)) = [x_{\ell n}, x_{hn}].$$

A task n therefore consists of the collection of menus

$$M_n(y) = \{p_n, \delta_y\},$$

with y varied over $[x_{\ell n}, x_{hn}]$ to elicit the certainty equivalent $\bar{y}_n$ of p_n. For a sure payoff y in task n, define its indirect value as

$$S_n(y; \alpha) = \max_{r \in [x_{\ell n}, y]} \left\{ v_{\alpha,\sigma}(r) + v_{\alpha,\sigma}(y - r) \right\}.$$

Let $V_n(\alpha, \lambda)$ denote the optimized value of the risky lottery. The certainty equivalent $\bar{y}_n(\alpha, \lambda)$ satisfies

$$S_n(\bar{y}_n(\alpha, \lambda); \alpha) = V_n(\alpha, \lambda).$$

If the risky lottery is evaluated with the no-loss reference point $x_{\ell n}$, the no-loss valuation is

$$V_n^{NL}(\alpha) = v_{\alpha,\sigma}(x_{\ell n}) + \pi_n v_{\alpha,\sigma}(D_n),$$

where $D_n = x_{hn} - x_{\ell n}$. Let $\bar{y}_n^{NL}(\alpha)$ denote the corresponding no-loss certainty equivalent, defined by $S_n(\bar{y}_n^{NL}(\alpha); \alpha) = V_n^{NL}(\alpha)$.

Step 1: A no-loss lottery identifies curvature. As shown in Section 3.3.2, sufficiently low dispersion can make aspiration unattractive. In particular, consider a lottery satisfying $x_{h1} \leq 2x_{\ell 1}$. Raising the reference point above $x_{\ell 1}$ cannot increase utility in the high state and weakly lowers utility in the low state. The lottery is therefore evaluated at $r = x_{\ell 1}$ for every $\alpha < 1$ and $\lambda \geq 0$. The

sure payoff at the certainty equivalent is also evaluated at $r = x_{\ell 1}$. Indifference then gives

$$\frac{v_\alpha(\bar{y}_1 - x_{\ell 1})}{v_\alpha(D_1)} = \pi_1 \tag{5}$$

where $D_1 = x_{h1} - x_{\ell 1}$. Since v_α is strictly increasing and strictly concave, any certainty equivalent generated by the model must satisfy[11]

$$x_{\ell 1} < \bar{y}_1 < x_{\ell 1} + \pi_1 D_1.$$

For a given observation in this interval, the ratio on the left-hand side of equation (5) is strictly decreasing in α so it uniquely determines the curvature.

Step 2: An aspirational lottery identifies loss sensitivity. Once α is known, consider a second lottery and compute its no-loss certainty equivalent $\bar{y}_2^{NL}(\alpha)$. If the observed certainty equivalent satisfies

$$\bar{y}_2 > \bar{y}_2^{NL}(\alpha),$$

the lottery must be evaluated with an aspirational reference point, since its value exceeds the no-loss benchmark. In the aspirational regime, the certainty equivalent is strictly decreasing in λ, so the observed certainty equivalent uniquely identifies loss sensitivity. Figure 4 illustrates both steps.

Proposition 6 (Global identification)**.** *Suppose $u = g = \ell = v_\alpha$, with $\alpha < 1$, $\lambda \geq 0$, and known $\sigma > 0$. Consider two exact certainty equivalents $(\bar{y}_1, \bar{y}_2)$ generated by the model. Suppose the first lottery satisfies $x_{h1} \leq 2x_{\ell 1}$, and let $\widehat{\alpha}$ be the unique solution to* (5)*. If $\bar{y}_2 > \bar{y}_2^{NL}(\widehat{\alpha})$, then (α, λ) is globally point identified.*

Observable restrictions. The model also restricts which certainty equivalents can be rationalized. The certainty equivalent from step 1 must satisfy the bound derived above. Once $\widehat{\alpha}$ is determined, let $\bar{y}_2^0(\widehat{\alpha})$ denote the certainty equivalent of the second lottery when the loss-sensitivity parameter equals zero. Since the no-loss reference point is always feasible and the lottery's value is weakly decreasing in λ, the second observation must satisfy

$$\bar{y}_2^{NL}(\widehat{\alpha}) \leq \bar{y}_2 \leq \bar{y}_2^0(\widehat{\alpha}).$$

An observation outside these bounds cannot be rationalized by any admissible parameter pair.

When the observed certainty equivalent equals the no-loss benchmark, it provides only a lower bound on loss sensitivity if the lottery induces some loss-sensitive agents to aspire, and no

[11]The lower bound follows from $v_\alpha(\bar{y}_1 - x_{\ell 1}) = \pi_1 v_\alpha(D_1) > 0$. The upper bound follows from strict concavity.

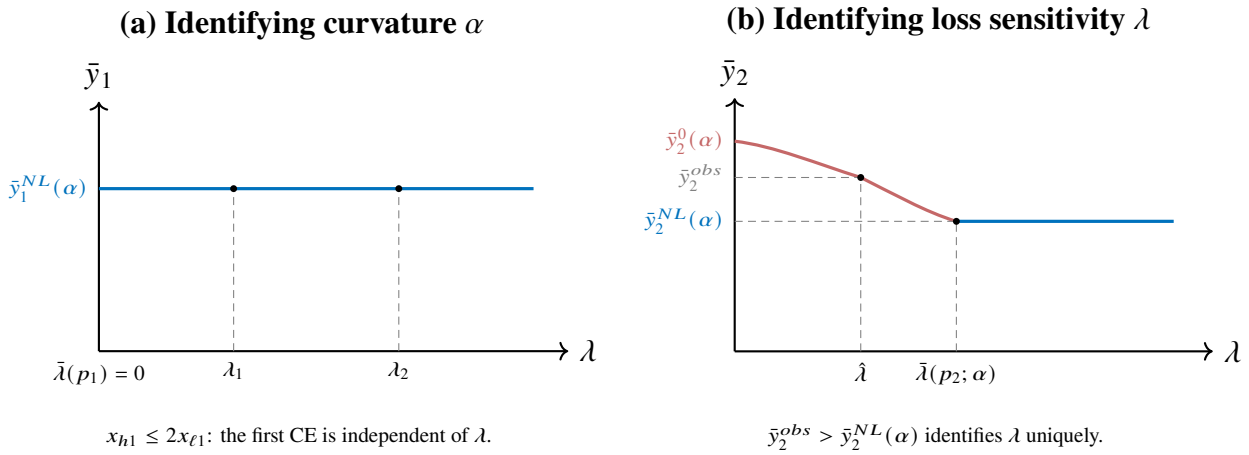


Figure 4: Identification from two certainty equivalents. Panel (a) uses a low-dispersion lottery that is evaluated in the no-loss regime for all admissible loss sensitivities, so its certainty equivalent identifies curvature α. Given $\widehat{\alpha}$, panel (b) uses a lottery that admits aspiration. Its certainty equivalent is strictly decreasing in λ above the no-loss benchmark and constant once the no-loss regime is optimal. An observed certainty equivalent strictly above the no-loss benchmark identifies λ uniquely. The curves are schematic.

information otherwise. However, for a fixed support with $A^*(x_\ell, x_h) > 0$, the cutoff satisfies

$$\bar{\lambda}(p_\pi) = \frac{\pi}{1-\pi} A^*(x_\ell, x_h) \longrightarrow \infty \qquad \text{as } \pi \to 1.$$

Thus, for any finite loss sensitivity, a sufficiently high probability of the high outcome makes aspiration optimal. The corresponding certainty equivalent then uniquely identifies loss sensitivity.

5 Applications: Reference-Point Adjustment

The previous sections fix a binary lottery and study how the agent chooses the reference point. I now let the agent choose an action that determines the lottery itself. Changes in preferences or economic opportunities can then affect behavior in two ways: directly through the action, and indirectly by changing the reference point, which in turn changes the incentives for the action.

I study this interaction in two applications corresponding to the two margins of a binary lottery. In portfolio choice, the action changes state-contingent payoffs while probabilities remain fixed. In effort choice, the action changes the probability of success while payoffs remain fixed. In each case, I compare the endogenous-reference problem with a benchmark in which the initially optimal reference point is held fixed. This comparison shows how much of the behavioral response comes from the adjustment of the reference point itself.

Before going to the applications, consider an interior optimum (a^*, r^*) of

$$\max_{a,r} F(a, r; z),$$

where a is a one-dimensional action and z is a preference parameter or a parameter of the economic environment. Suppose the Hessian of F with respect to (a, r) is negative definite. Holding the reference point fixed at r^*, the local response of the action is

$$\left.\frac{\partial a^{\text{fix}}}{\partial z}\right|_{r=r^*} = -\frac{F_{az}}{F_{aa}}.$$

When the reference point can also adjust,

$$\frac{da^*}{dz} = \left.\frac{\partial a^{\text{fix}}}{\partial z}\right|_{r=r^*} - \frac{F_{ar}}{F_{aa}}\frac{dr^*}{dz}. \tag{6}$$

Since $F_{aa} < 0$, the reference-point adjustment term has the sign of $F_{ar}dr^*/dz$. It reinforces the fixed-reference response when the two have the same sign and works against it otherwise. The applications below show that this second channel can either reinforce or offset the first.

5.1 Portfolio Choice: Changing the Payoffs

In the portfolio application, an investor allocates wealth across two equally likely states by trading Arrow–Debreu securities. Each security pays one unit in one state and nothing in the other. This is the two-state portfolio environment studied experimentally by Choi et al. (2007). The investor chooses both the state-contingent payoffs and a wealth target, while probabilities remain fixed. I study how relative prices affect the portfolio and the chosen reference point, and then examine how loss sensitivity affects regime selection and portfolio insurance.

Environment. Let $p_1, p_2 > 0$ denote the prices of the two securities and $\rho = \log(p_1/p_2)$ their log price ratio. I focus on $\rho < 0$; the case $\rho > 0$ is symmetric. State 1 is then cheaper, so write its payoff as x_H and the payoff in state 2 as x_L, with $x_H \geq x_L$. I normalize the budget so that the certain portfolio (m, m) is feasible at every price ratio:

$$e^{\rho}x_H + x_L = m(1 + e^{\rho}). \tag{7}$$

A fall in ρ therefore lowers the relative price of the high-state security and pivots the budget line around (m, m). As in Assumption 2, the feasible reference-point set is the range of outcomes

attainable on the budget set. Thus

$$R(\rho, m) = [0, m(1 + e^{-\rho})],$$

and the investor jointly chooses the portfolio and reference point:

$$\begin{aligned} \max_{\substack{x_H, x_L \geq 0 \\ r \in R(\rho, m)}} \quad & \frac{1}{2}U(x_H, r; \lambda) + \frac{1}{2}U(x_L, r; \lambda) \\ \text{s.t.} \quad & e^{\rho} x_H + x_L = m(1 + e^{\rho}). \end{aligned} \tag{8}$$

Reference points above x_H can be ignored. A reference point is therefore no-loss when $r \leq x_L$ and aspirational when $x_L < r \leq x_H$. I measure relative demand by

$$s := \frac{x_H}{x_H + x_L}.$$

Moving s toward $1/2$ reduces the payoff spread and provides greater insurance to the low-payoff state. At an optimum in regime $j \in \{NL, A\}$, where NL denotes the no-loss regime and A the aspirational regime, write (x_H^j, x_L^j, r^j) for the optimal choices and s^j for the corresponding relative demand. In the fixed-reference benchmark, the investor reoptimizes the portfolio while holding the reference point at its initially optimal value r^j; write x_L^{fix} and s^{fix} for the resulting low-state payoff and relative demand.

5.1.1 Prices

For the price comparison, I maintain the shifted-power specification for g with $0 < \alpha_g < 1$. At an aspirational optimum, additionally suppose

$$-\frac{g'(x_H^A - r^A)}{g''(x_H^A - r^A)} \geq x_H^A - m, \tag{9}$$

which ensures that the matched fixed-reference investor shifts wealth toward the cheaper state when ρ falls.

Proposition 7 (Prices and reference-point adjustment)**.** *Fix $\rho < 0$ and suppose g has the shifted-power form with $0 < \alpha_g < 1$. In the aspirational regime, consider a stable interior optimum, with a negative-definite Hessian in (x_L, r), and suppose* (9) *holds.*

(i) At an interior no-loss optimum, a fall in ρ lowers the reference point, raises x_H, and lowers x_L.

(ii) At a stable interior aspirational optimum, a fall in ρ instead raises the reference point and x_H, lowers x_L, and widens the shortfall $r^A - x_L^A$.

In both regimes, reference-point adjustment amplifies the local portfolio response:

$$\frac{ds^j}{d\rho} < \left.\frac{\partial s^{\text{fix}}}{\partial \rho}\right|_{r=r^j} < 0, \qquad j \in \{NL, A\}.$$

When the high-state security becomes relatively cheaper, the investor raises x_H and lowers x_L. The reference point responds differently across regimes. In the no-loss regime, the fall in x_L reduces the surplus above the reference point in the low state, increasing its marginal utility. This favors a lower reference point, while the rise in x_H works in the opposite direction. Under the maintained curvature condition, the first effect dominates, so r falls. Lowering the reference point increases surplus in both states, reducing their marginal utilities, but proportionally more in the low state. The investor therefore shifts additional wealth toward the cheaper high-state security.

In the aspirational regime, the reference point instead rises. The increase in x_H reduces the marginal utility of surplus above the reference point, making a higher aspiration more attractive. At the same time, the fall in x_L increases the shortfall and, by concavity of ℓ, weakly reduces its marginal cost. Both effects increase the attractiveness of a higher reference point. Raising the reference point then increases the marginal utility of high-state wealth and weakly reduces that of low-state wealth, reinforcing the shift toward the cheaper state.

At equal prices, the optimal portfolio is symmetric and the reference point is no-loss. Under the regularity condition in the next corollary, the reference point remains no-loss for prices sufficiently close to equal.

Corollary 4 (No-loss demand around equal prices)**.** *At $\rho = 0$,*

$$s(0, m, \lambda) = \frac{1}{2}$$

for every $\lambda \geq 0$. If the no-loss problem at (m, m) has a unique regular solution $r_m < m$, there exists some $\varepsilon > 0$, such that the optimal reference point is no-loss for every $|\rho| < \varepsilon$. On this interval, portfolio demand is independent of λ.

Thus small price differences can affect portfolio demand without loss sensitivity affecting it when the investor keeps a no-loss reference point.

5.1.2 Loss sensitivity

I next hold investment opportunities fixed and vary loss sensitivity. Let $V^{NL}(\rho, m)$ denote the maximum value over no-loss choices, $r \leq x_L$, and $V^A(\rho, m, \lambda)$ the supremum over aspirational choices, $x_L < r \leq x_H$. The no-loss value does not depend on λ, while the aspirational value is weakly decreasing in λ. Define the set of price environments in which aspiration is strictly preferred

by

$$\mathcal{A}(\lambda, m) := \left\{\rho : V^A(\rho, m, \lambda) > V^{NL}(\rho, m)\right\}.$$

Under the regularity condition of Corollary 4, $\mathcal{A}(\lambda, m)$ excludes a neighborhood of equal prices.

Proposition 8 (Loss sensitivity and regime selection)**.** *Fix (ρ, m) and consider two otherwise identical investors with $0 < \lambda_1 < \lambda_2$. The more loss-sensitive investor chooses a weakly smaller shortfall:*

$$\left(r^*(\rho, m, \lambda_2) - x_L^*(\rho, m, \lambda_2)\right)_+ \leq \left(r^*(\rho, m, \lambda_1) - x_L^*(\rho, m, \lambda_1)\right)_+.$$

Moreover, $\mathcal{A}(\lambda_2, m) \subseteq \mathcal{A}(\lambda_1, m)$.

Once a no-loss reference point is selected, further increases in λ do not affect the portfolio.

Figure 5 illustrates Proposition 8 using simulations with shifted CRRA utility. For each price ratio, the investor jointly chooses the portfolio and the reference point. The less loss-sensitive investor chooses an aspirational reference point for a wider range of price ratios. Demand can jump where the optimal regime switches, and these jumps need not be unique.

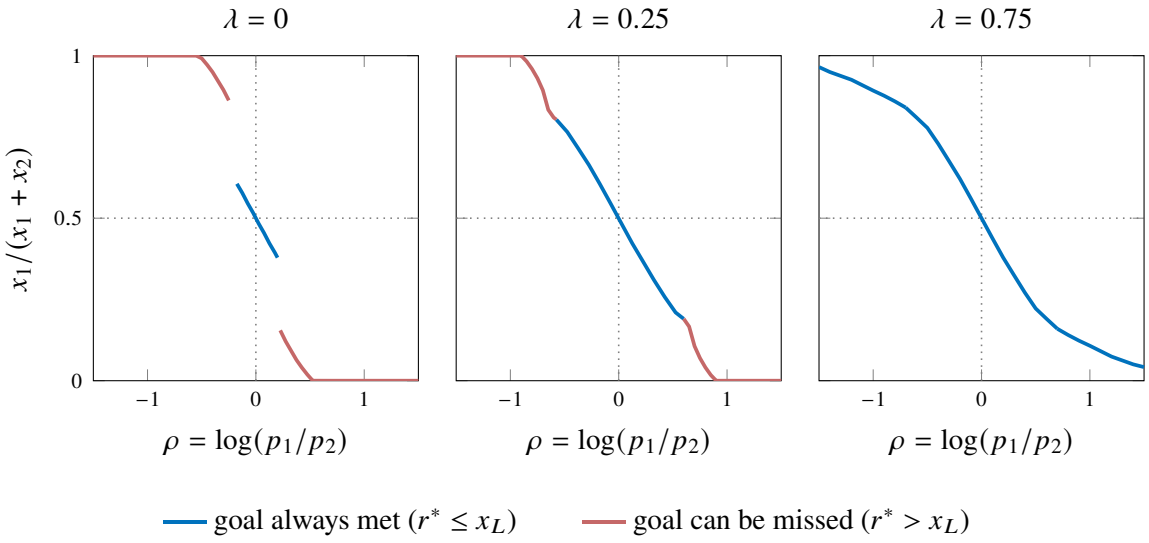


Figure 5: Simulated portfolio demand and loss sensitivity. Each panel plots relative demand for the state-1 security, $x_1/(x_1 + x_2)$, against the log price ratio ρ; for $\rho < 0$ this equals s. Simulations use the shifted CRRA specification of Assumption 3 with $\alpha_u = \alpha_g = \alpha_\ell = 0.25$, $\sigma = 1$ and $m = 0.75$. At $\lambda = 0.75$, the optimal reference point equals the low payoff for $|\rho| > 0.7$.

At high loss sensitivity, the investor chooses a no-loss reference point that is always met, and demand varies smoothly with prices. Smooth demand therefore does not imply that the investor is insensitive to losses: a loss-sensitive investor may simply choose an aspiration that is never missed.

Figure 5 also illustrates how the model can generate heterogeneous responses to relative prices, similar to those documented by Choi et al. (2007). For some investors, portfolio demand changes

smoothly with prices; for others, switches in the optimal reference-point regime can produce sharp changes in demand.

Proposition 8 shows that a more loss-sensitive investor accepts a smaller shortfall, but not how: the shortfall can shrink because x_L rises or because r falls. At an interior aspirational optimum, both happen and reinforce each other. The problem is the substitute case of Lemma A.3 with $a = x_L$, since $F^A_{x_L\lambda} > 0$, $F^A_{r\lambda} < 0$ and $F^A_{x_L r} < 0$. The fall in the reference point therefore strengthens the direct increase in insurance to the low-payoff state.

Proposition 9 (Loss sensitivity and portfolio insurance)**.** *Fix $\rho < 0$ and $m > 0$. While a no-loss optimum remains selected, the portfolio and reference point are independent of λ.*

At a stable interior aspirational optimum,

$$\frac{dx_L^A}{d\lambda} > \left.\frac{\partial x_L^{\text{fix}}}{\partial \lambda}\right|_{r=r^A} > 0, \qquad \frac{dx_H^A}{d\lambda} < 0, \qquad \frac{dr^A}{d\lambda} < 0.$$

Hence the payoff spread and the shortfall both decrease, and the portfolio moves toward equal state-contingent payoffs. Moreover,

$$\frac{ds^A}{d\lambda} < \left.\frac{\partial s^{\text{fix}}}{\partial \lambda}\right|_{r=r^A} < 0.$$

A more loss-sensitive investor increases wealth in the low-payoff state and chooses a lower reference point. These adjustments reinforce one another because $F_{x_L r} < 0$: lowering the reference point increases the marginal return to low-state wealth. Hence, the increase in insurance is larger than when the reference point is held fixed.

5.2 Effort Choice: Changing the Probability of Success

I show that when the chosen aspiration is sufficiently close to the low outcome, greater loss sensitivity reduces effort, holding productivity fixed. Under a parametric specification, aspirations become optimal continuously as productivity increases, so the negative effort response occurs just above the threshold of productivity at which aspirations become optimal. I also show that loss sensitivity leaves effort unchanged at low productivity, reduces it at intermediate productivity, and increases it at high productivity. Allowing aspirations to adjust can therefore reverse the positive effort response obtained when the aspiration is held fixed.

Environment. An agent chooses effort $e \geq 0$ and a reference point $r \in [x_\ell, x_h]$. The environment can be interpreted as a student choosing how hard to study and a target level of performance.

High performance x_h occurs with probability $qs(e)$, while low performance $x_\ell < x_h$ occurs with probability $1 - qs(e)$. The parameter $q \in (0, 1]$ measures productivity.

The agent jointly chooses effort and the reference point to solve

$$\max_{\substack{e\geq 0 \\ r\in[x_\ell, x_h]}} F(r, e; q, \lambda) = qs(e)H(r) + [1 - qs(e)]L(r, \lambda) - c(e), \tag{10}$$

where

$$H(r) = u(r) + g(x_h - r), \qquad L(r, \lambda) = u(x_\ell) - \lambda\ell(r - x_\ell).$$

Here, $H(r)$ and $L(r, \lambda)$ denote utility in the high- and low-outcome states, respectively. Maintain Assumption 1. Suppose $s(0) = 0$, $s' > 0$, $s'' < 0$, and $s(e) < 1$, while $c' > 0$ and $c'' \geq 0$. The reference point is no-loss when $r = x_\ell$ and aspirational when $r > x_\ell$. At a no-loss reference point, loss sensitivity does not affect the objective.

At an interior aspirational optimum, the first-order conditions for effort and the reference point are

$$qs'(e)[H(r) - L(r, \lambda)] = c'(e), \tag{11}$$

$$qs(e)H'(r) = [1 - qs(e)]\lambda\ell'(r - x_\ell). \tag{12}$$

The effort condition equates the marginal cost of effort with the marginal increase in the probability of success, weighted by the utility difference between success and failure. The reference-point condition balances the expected marginal gain from raising the aspiration when it is attained against the expected marginal cost of a larger shortfall when it is missed.

To characterize the fixed-reference response, define $\phi(e) := c'(e)/s'(e)$, which is strictly increasing under the maintained assumptions. The effort first-order condition becomes

$$\phi(e) = q[H(r) - L(r, \lambda)].$$

Holding an aspirational reference point fixed, greater loss sensitivity increases the utility difference between success and failure. Differentiating gives

$$\left.\frac{\partial e}{\partial \lambda}\right|_r = \frac{q\ell(r - x_\ell)}{\phi'(e)} > 0. \tag{13}$$

When the agent can also adjust the aspiration, greater loss sensitivity makes raising it less attractive at a given level of effort ($F_{r\lambda} < 0$). This matters because at an interior aspirational

optimum with $\lambda > 0$, effort and the reference point are local complements:

$$F_{er} = qs'(e)\big[H'(r) + \lambda \ell'(r - x_\ell)\big] > 0,$$

where the sign follows from the reference-point first-order condition. Thus, a lower aspiration reduces the marginal return to effort and can offset the positive fixed-reference response.

To determine which effect dominates, let $p^* := qs(e^*)$. At a stable interior aspirational optimum, differentiating both first-order conditions gives

$$\operatorname{sgn}\left(\frac{de^*}{d\lambda}\right) = \operatorname{sgn}\left[\underbrace{-F_{rr}\ell(r^* - x_\ell)}_{\text{failure avoidance}} \underbrace{- \frac{1 - p^*}{p^*}\lambda\big[\ell'(r^* - x_\ell)\big]^2}_{\text{reference-point adjustment}}\right]. \tag{14}$$

The first term is positive and captures the incentive to exert more effort to avoid failure. The second is negative and captures the effect of reference-point adjustment. As the aspiration approaches the low outcome, $\ell(r^* - x_\ell)$ tends to zero, while $\ell'(r^* - x_\ell)$ stays positive. The negative effect can therefore dominate when the chosen aspiration is sufficiently close to the low outcome.

Proposition 10 (Effort response near the low aspiration)**.** *Fix $\lambda_0 > 0$ and suppose $H'(x_\ell) > 0$. Under the maintained smoothness assumptions, suppose the Hessian of F with respect to (e, r) is negative definite at an interior aspirational optimum. Then there exists $\varepsilon > 0$ such that, whenever $0 < r^* - x_\ell < \varepsilon$,*

$$\left.\frac{de^*}{d\lambda}\right|_{q,\lambda=\lambda_0} < 0.$$

The proposition shows that greater loss sensitivity can reduce effort without imposing a particular parametric specification of utility or effort. In the parametric specification below, aspirations become optimal continuously from x_ℓ, so the negative effort response occurs just above the activation threshold.

A parametric specification. To characterize how the two forces vary with productivity, consider the case $x_\ell = 0$ with

$$\begin{aligned} u = g = \ell = v_{\alpha,\sigma}, \qquad s(e) = 1 - e^{-\beta e}, \qquad c(e) = \kappa e, \\ v_{\alpha,\sigma}(z) = \begin{cases} \dfrac{(z+\sigma)^\alpha - \sigma^\alpha}{\alpha}, & \alpha \neq 0, \\ \log\left(\dfrac{z+\sigma}{\sigma}\right), & \alpha = 0, \end{cases} \qquad \alpha < 1, \quad \sigma, \beta, \kappa > 0. \end{aligned} \tag{15}$$

The parameter β determines the effectiveness of effort since a higher β raises the probability of

getting the high outcome for a given level of effort. The parameter κ corresponds to the marginal cost of effort.

To determine when the negative effort response established above arises, I first characterize how the chosen aspiration varies with productivity. After optimizing over effort, let $q(r;\lambda)$ denote the productivity at which an interior aspiration r satisfies the reference-point first-order condition. Holding effort fixed, a higher aspiration requires a higher probability of success, but a higher aspiration also induces more effort, which raises the probability of success at a given productivity. These effects work in opposite directions. However, Appendix A.8 shows that, for each $\lambda > 0$, there is a finite bound $\bar{\beta}(\lambda)$ such that $q(r;\lambda)$ is strictly increasing in r when $\beta > \bar{\beta}(\lambda)$. Under this condition, the aspirational optimum is unique and the chosen aspiration increases with productivity.

The effect of loss sensitivity on effort can then be characterized by two productivity thresholds. Define $q^A(\lambda) := q(0;\lambda)$ as the productivity at which aspiration becomes optimal, and let $q^E(\lambda)$ denote the productivity at which the effect of loss sensitivity on effort changes sign.

Theorem 2 (Three productivity regimes)**.** *Fix $\lambda > 0$. Under* (15) *and $\beta > \bar{\beta}(\lambda)$, there exist unique thresholds*

$$0 < q^A(\lambda) < q^E(\lambda) < 1$$

such that

$$\frac{de^*}{d\lambda}\begin{cases} = 0, & q < q^A(\lambda), \\ < 0, & q^A(\lambda) < q < q^E(\lambda), \\ = 0, & q = q^E(\lambda), \\ > 0, & q^E(\lambda) < q \leq 1. \end{cases} \tag{16}$$

Below $q^A(\lambda)$, the optimal reference point is no-loss. Above $q^A(\lambda)$, it is aspirational and increases strictly with productivity. Along the aspirational branch, $\frac{dr^*}{d\lambda} < 0$.

Figure 6 illustrates how the effect of loss sensitivity on effort varies across productivity levels. Consider two students who only differ slightly in loss sensitivity and face the same productivity. At low productivity, both choose not to aspire, so loss sensitivity does not affect their effort. Just above the activation threshold $q^A(\lambda)$, the chosen aspiration is still close to the low outcome. The penalty from failing is small, so the more loss-sensitive student has little incentive to exert effort. However, greater loss sensitivity increases the marginal cost of raising the aspiration since $\ell'(0) > 0$. The more loss-sensitive student chooses a lower aspiration, which reduces the marginal return to effort because effort and aspirations are complements. When their aspirations are low, this effect dominates, so the more loss-sensitive student aspire lower and produces less effort than the less loss-sensitive student. At higher productivity, aspirations are higher and failure becomes more costly.

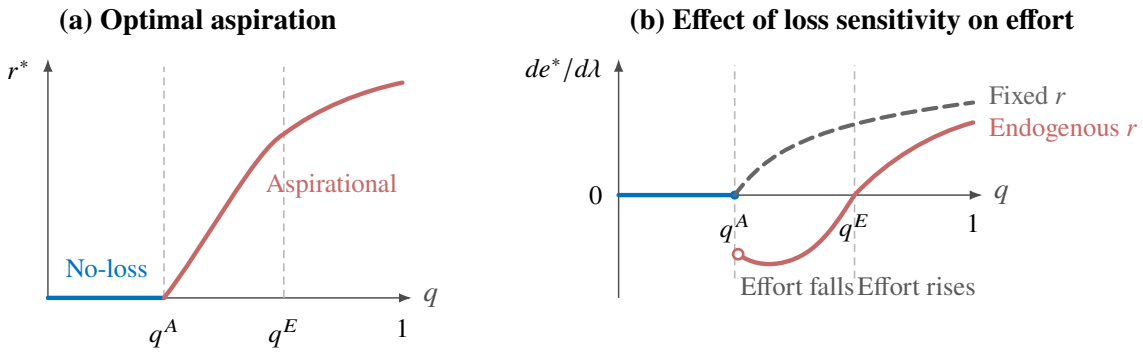


Figure 6: Effort and aspiration adjustment across productivity (schematic). Panel (a) shows continuous activation of the optimal aspiration at q^A. Panel (b) shows that the endogenous-reference response to loss sensitivity is zero below activation, negative between q^A and q^E, and positive above q^E. The curves just illustrate the sign of the derivative, not the magnitudes.

The incentive to avoid failure eventually dominates aspiration adjustment, so the more loss-sensitive student exerts more effort above $q^E(\lambda)$.

With a fixed aspirational reference point, greater loss sensitivity raises effort at an interior optimum. When the reference point can adjust, it can instead leave effort unchanged, reduce it, or increase it, depending on productivity.[12]

Corollary 5 (Productivity thresholds)**.** *Under the conditions of Theorem 2, both $q^A(\lambda)$ and $q^E(\lambda)$ decrease with β and increase with κ. In addition, $q^A(\lambda)$ increases with λ.*

These comparative statics show how the effectiveness and cost of effort affect the two productivity thresholds. When effort is more effective, aspiration becomes optimal at lower productivity and the positive effort response to loss sensitivity appears sooner. When effort is more costly, both transitions require higher productivity. Kim (2023) also obtains complementarity between effort and the reference point. Here, I instead study how productivity changes the effect of loss sensitivity on effort.

The two applications differ in how the action and the reference point interact. At an interior aspirational portfolio optimum, low-state wealth and the reference point are local substitutes, so lowering the reference point reinforces the insurance response to loss sensitivity. In effort choice, effort and the reference point are local complements, so a lower aspiration works against the direct incentive to increase effort and can reverse its response to loss sensitivity.

[12]The comparative statics are local in λ, holding productivity fixed. The restriction on β is sufficient but not necessary: it guarantees a unique aspirational branch, continuous activation, and $q^E(\lambda) < 1$.

6 Conclusion

This paper studies how people choose their aspirations and how these choices influence decisions under uncertainty. Choosing a more demanding aspiration involves a trade-off. On the one hand, it can be more rewarding when it is attained. On the other hand, it reduces the gain from exceeding it and increases the likelihood and cost of falling short. I develop a parsimonious model to characterize this trade-off. The utility function distinguishes the value of attaining an aspiration, the gain from exceeding it, and the penalty from falling short. A loss-sensitivity parameter parametrizes this trade-off by scaling the penalty for falling short without changing the reward from attainment. By varying this parameter, I can characterize when individuals find it worthwhile to choose aspirations they may fail to attain, and how high they choose to aspire. I then study how these choices interact with actions that affect the distribution of outcomes.

For any finite lottery, I show that there exists a cutoff in loss sensitivity that separates two regimes of aspiration choice. Below the cutoff, the agent chooses an aspiration that can be missed; above it, she chooses a reference point attained in every state. The cutoff depends on the lottery. I specialize to binary lotteries to characterize how this cutoff depends on the lottery's properties. A higher probability of the high outcome or greater payoff dispersion can make aspiration more attractive, so the same individual may aspire in one environment but not another. This also generates a plateau in certainty equivalents. Once a no-loss reference point is optimal, further increases in loss sensitivity do not affect valuation, even though outcomes are uncertain. Two individuals with different loss sensitivities can therefore value one risky lottery equally and another differently.

The model generates a distinction between aspirational reference points, which can be missed, and no-loss reference points, which are attained in every state. For any finite lottery, a cutoff in loss sensitivity determines which regime is optimal. Changes in success probabilities and payoff dispersion affect this choice, so the same individual can aspire in one environment and choose a no-loss reference point in another. In the latter regime, loss sensitivity no longer affects valuation, even though outcomes remain uncertain. A more loss-sensitive individual lies above the cutoff for more lotteries. The individuals who are most sensitive to falling short are therefore those whose loss sensitivity is irrelevant for valuation in the widest range of environments: they avoid the penalty by choosing aspirations they always attain.

When the agent can also take an action that affects her prospects, the action and the aspiration interact. The action changes the distribution of outcomes, while the aspiration affects the incentives to act. I compare their joint choice with a benchmark in which the initially optimal aspiration is held fixed. In portfolio choice, when a state becomes relatively cheaper, the investor shifts more wealth toward it when her wealth target can adjust than when it is held fixed. At an interior aspirational optimum, aspiration adjustment also reinforces the shift toward insurance when loss sensitivity

increases. In effort choice, aspiration adjustment can instead work against the direct incentive to avoid failure. A more loss-sensitive student may lower her aspiration, reducing the incentive to exert effort. Under a parametric specification, this adjustment can reverse the positive effect of loss sensitivity obtained when the aspiration is held fixed. Allowing aspirations to adjust can therefore change not only the magnitude of behavioral responses, but also their direction.

The optimal choice of aspirations also provides a way to identify preferences when reference points are unobserved. In general, it is difficult to distinguish whether differences in observed choices come from preferences or from the reference points used to evaluate outcomes. In my model, although aspirations can differ across lotteries, they must be optimal under the same preferences. This restricts how they can vary across choice problems. I use this restriction together with a parametric specification to show that two certainty equivalents can identify curvature and loss sensitivity. A sufficiently low-dispersion lottery identifies curvature because it is always evaluated in the no-loss regime. Given this curvature, a second lottery whose certainty equivalent exceeds its no-loss benchmark identifies loss sensitivity. Thus, preferences can be recovered without observing the aspirations the agent chooses.

More generally, the paper shows that individuals can respond to changes in preferences and economic opportunities by adjusting the aspirations they choose. When failure becomes more costly, they may lower their aspirations rather than simply change their actions. This has two implications. First, individuals may lower their aspirations enough that every possible outcome is evaluated above the reference point. In this case, preferences governing how individuals evaluate losses or failure become behaviorally irrelevant, even though outcomes are still uncertain. Second, when individuals also choose actions that affect their prospects, adjusting their aspirations can reinforce or offset the direct response to changes in preferences and economic opportunities. Taking aspirations as fixed can therefore lead to different predictions about behavior.

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A Proofs

A.1 Choice Problem

Lemma A.1 (Existence)**.** *Maintain Assumption 1 and Assumption 2. For every finite menu M, the joint problem*

$$\max_{p\in M,\, r\in\mathcal{R}(M)} \mathbb{E}_p\big[U(X,r)\big]$$

admits a solution (p^*_M, r^*_M).

Proof of Lemma A.1. By Assumption 2, $\mathcal{R}(M) = [\underline{x}(M), \overline{x}(M)]$ is nonempty and compact. Fix $x \in \mathcal{X}$. By assumption 1, the utility $U(x,r)$ is continuous in r on $\mathcal{R}(M)$: it is continuous on $\{r < x\}$ and on $\{r > x\}$, and at $r = x$, since $g(0) = \ell(0) = 0$ it is also continuous at $r = x$.

For any $p \in \Delta(\mathcal{X})$, the expected utility $\mathbb{E}_p\big[U(X,r)\big] = \sum_{x\in\mathcal{X}} p(x)\,U(x,r)$ is a finite sum of continuous functions so it is continuous on $\mathcal{R}(M)$. By Weierstrass' theorem, for each $p \in M$ the set of optimal reference points for lottery p

$$\arg\max_{r\in\mathcal{R}(M)} \mathbb{E}_p\big[U(X,r)\big]$$

is nonempty, and the indirect value

$$V_M(p) := \max_{r\in\mathcal{R}(M)} \mathbb{E}_p\big[U(X,r)\big]$$

is well-defined. Since M is finite, $\max_{p\in M} V_M(p)$ is attained at some $p^*_M \in M$. Choose any $r^*_M \in \arg\max_{r\in\mathcal{R}(M)} \mathbb{E}_{p^*_M}[U(X,r)]$. Then (p^*_M, r^*_M) is a solution of the problem. ∎

A.2 Within-lottery reference point choice

A.2.1 Proof of Lemma 1

Proof. Fix a menu M and a lottery p. For any $r \in \mathcal{R}^{NL}_M(p) = [\underline{x}(M), x_\ell(p)]$, every outcome satisfies $x \geq r$, so

$$\mathbb{E}_p[U(X,r)] = u(r) + \mathbb{E}_p[g(X-r)]$$

which is independent of λ and strictly concave in r by Assumption 1. Since $\mathcal{R}^{NL}_M(p)$ is compact, it admits a unique maximizer $r^{NL}_M(p)$. ∎

A.2.2 Proof of Theorem 1

In the next lemma, I show that for sufficiently high loss sensitivity, no aspirational reference point does better than $x_\ell(p)$.

Lemma A.2 (Aspirations collapse to the lower outcome at high loss sensitivity)**.** *Fix a nondegenerate lottery p, and let $x_\ell := \min \operatorname{supp}(p) < x_h := \max \operatorname{supp}(p)$. Define*

$$V^A(p,\lambda) := \sup_{r\in(x_\ell,x_h]} \mathbb{E}_p[U(X,r)].$$

Then there exists $\lambda_0(p) < \infty$ such that for all $\lambda \geq \lambda_0(p)$,

$$V^A(p,\lambda) = \lim_{r\downarrow x_\ell} \mathbb{E}_p[U(X,r)] = \mathbb{E}_p[U(X,x_\ell)].$$

Proof. Fix a nondegenerate lottery p with finite support, and write $x_\ell := \min \operatorname{supp}(p)$, $x_h := \max \operatorname{supp}(p)$. The probability of the lowest outcome is strictly positive $p(x_\ell) > 0$. Between consecutive outcomes in the support

$$\frac{d}{dr}\,\mathbb{E}_p[U(X,r)] = \sum_{x\geq r} p(x)\big[u'(r) - g'(x-r)\big] - \lambda \sum_{x<r} p(x)\ell'(r-x).$$

For $r > x_\ell$, assumption 1 implies $\ell'(r-x_\ell) \geq \ell'(x_h - x_\ell)$ and

$$\frac{d}{dr}\,\mathbb{E}_p[U(X,r)] \leq (1-p(x_\ell))u'(x_\ell) - \lambda p(x_\ell)\ell'(x_h-x_\ell).$$

Since $\ell'(x_h - x_\ell) > 0$, define

$$\lambda_0(p) := \frac{(1-p(x_\ell))u'(x_\ell)}{p(x_\ell)\ell'(x_h-x_\ell)} < \infty.$$

For $\lambda \geq \lambda_0(p)$, the derivative is nonpositive whenever it exists. Since the objective is continuous in r, it is therefore nonincreasing on $[x_\ell, x_h]$. Hence

$$V^A(p,\lambda) = \sup_{r\in(x_\ell,x_h]} \mathbb{E}_p[U(X,r)] = \lim_{r\downarrow x_\ell} \mathbb{E}_p[U(X,r)] = \mathbb{E}_p[U(X,x_\ell)] \quad \forall \lambda \geq \lambda_0$$

■

Proof of Theorem 1. Fix a menu M and a nondegenerate lottery $p \in M$, and write $x_\ell := x_\ell(p)$ and $x_h := x_h(p)$. On the no-loss region, losses never happen, so

$$\mathbb{E}_p[U(X,r)] = u(r) + \mathbb{E}_p[g(X-r)],$$

and therefore $V_M^{NL}(p)$ is independent of λ. For every aspirational $r \in (x_\ell, x_h]$,

$$\mathbb{E}_p[U(X,r)] = G_p(r) - \lambda L_p(r),$$

where $L_p(r) := \sum_{x<r} p(x)\ell(r-x) > 0$. Hence $V^A(p,\lambda)$ is the supremum of strictly decreasing affine functions of λ, so it is convex and weakly decreasing.

To show which regime is optimal at the cutoff itself, I next show that $V^A(p,\lambda)$ is continuous in λ. Note that $L_p(r) \leq \ell(x_h - x_\ell)$ for every $r \in (x_\ell, x_h]$. Thus, for $0 \leq \lambda_1 < \lambda_2$,

$$G_p(r) - \lambda_1 L_p(r) - (\lambda_2 - \lambda_1)\ell(x_h - x_\ell) \leq G_p(r) - \lambda_2 L_p(r) \leq G_p(r) - \lambda_1 L_p(r).$$

Taking the supremum over $r \in (x_\ell, x_h]$ gives

$$0 \le V^A(p,\lambda_1) - V^A(p,\lambda_2) \le (\lambda_2 - \lambda_1)\ell(x_h - x_\ell).$$

so $V^A(p,\lambda)$ is continuous on $[0,\infty)$.

By Lemma A.2, there exists $\lambda_0(p) < \infty$ such that, for every $\lambda \ge \lambda_0(p)$,

$$V^A(p,\lambda) = \mathbb{E}_p[U(X, x_\ell)] \le V_M^{NL}(p),$$

where the inequality follows because x_ℓ is feasible in the no-loss region. Therefore the cutoff at which agents stop aspiring is finite:

$$\bar{\lambda}_M(p) := \inf\{\lambda \ge 0 : V^A(p,\lambda) \le V_M^{NL}(p)\} \le \lambda_0(p) < \infty.$$

I now characterize the optimal reference point for each $\lambda \ge 0$. By definition of $\bar{\lambda}_M(p)$, if $\lambda < \bar{\lambda}_M(p)$ then $V^A(p,\lambda) > V_M^{NL}(p)$, so every optimal reference point is aspirational. By continuity, the set $\{\lambda \ge 0 : V^A(p,\lambda) \le V_M^{NL}(p)\}$ is closed, so $\bar{\lambda}_M(p)$ belongs to the set. Hence $V^A(p,\bar{\lambda}_M(p)) \le V_M^{NL}(p)$, and $r_M^{NL}(p)$ is optimal at the cutoff. If $\bar{\lambda}_M(p) > 0$, the inequality holds with equality, so an aspirational reference point may also tie. Finally, if $\lambda > \bar{\lambda}_M(p)$, then for every aspirational r, $G_p(r) - \lambda L_p(r) < G_p(r) - \bar{\lambda}_M(p)L_p(r) \le V^A(p,\bar{\lambda}_M(p)) \le V_M^{NL}(p)$, where the strict inequality follows from $L_p(r) > 0$. Hence no aspirational reference point is optimal. By Lemma 1, the unique optimum is $r_M^{NL}(p)$. ■

A.3 Proofs for Binary lotteries (section 3)

Throughout this section, I maintain Assumptions 1, 2, and 3.

A.3.1 Proof of Lemma 2

Proof. I show that the binary reference-point expected utility is single-peaked under assumption 3. This implies that whenever an aspirational reference point is optimal, the aspirational maximizer is unique. It is sufficient to show that the second order condition of r changes sign at most once, and only from positive to negative under assumption 3.

Let $F(r;p,\lambda) := \mathbb{E}_p[U(X,r)]$, write $d := x_h - x_\ell$, and let $t := r - x_\ell \in (0,d]$. On the aspirational region,

$$F_{rr}(x_\ell + t; p, \lambda) = \pi\big[u''(x_\ell + t) + g''(d - t)\big] - (1-\pi)\lambda\ell''(t).$$

Since $\ell'' < 0$, this can be written as

$$F_{rr}(x_\ell + t; p, \lambda) = -(1-\pi)\ell''(t)\big[\lambda - B_p(t)\big],$$

where

$$B_p(t) := \frac{\pi}{1-\pi}\frac{-u''(x_\ell + t) - g''(d-t)}{-\ell''(t)}.$$

It is therefore enough to show that $B_p(t)$ is increasing.

Under Assumption 3, let $a_j := 2 - \alpha_j > 1$ and rewrite

$$B_p(t) = \frac{\pi}{1-\pi}\left[C_u \frac{(t+\sigma)^{a_\ell}}{(x_\ell + t + \sigma)^{a_u}} + C_g \frac{(t+\sigma)^{a_\ell}}{(d-t+\sigma)^{a_g}}\right].$$

where

$$C_u := \frac{1-\alpha_u}{1-\alpha_\ell} > 0, \qquad C_g := \frac{1-\alpha_g}{1-\alpha_\ell} > 0.$$

For the first term,

$$\frac{d}{dt}\log\frac{(t+\sigma)^{a_\ell}}{(x_\ell + t + \sigma)^{a_u}} = \frac{a_\ell}{t+\sigma} - \frac{a_u}{x_\ell + t + \sigma} > 0,$$

since $a_\ell \geq a_u$. For the second,

$$\frac{d}{dt}\log\frac{(t+\sigma)^{a_\ell}}{(d-t+\sigma)^{a_g}} = \frac{a_\ell}{t+\sigma} + \frac{a_g}{d-t+\sigma} > 0.$$

Thus $B_p(t)$ is strictly increasing, so F_{rr} can change sign at most once, from positive to negative. It follows that F_r can cross zero from positive to negative at most once, and F has at most one local maximum on $(x_\ell, x_h]$. ∎

A.3.2 Proof of Proposition 2

Proof. Let $F(r; p, \lambda) := \mathbb{E}_p[U(X, r)]$. For any aspirational reference point $r > x_\ell$,

$$F(r; p, \lambda) - F(x_\ell; p) = (1-\pi)\ell(r - x_\ell)\left[\frac{\pi}{1-\pi}\frac{u(r) + g(x_h - r) - u(x_\ell) - g(x_h - x_\ell)}{\ell(r - x_\ell)} - \lambda\right].$$

Suppose first that $A^*(p) \leq 0$. Then the term in brackets is nonpositive for every $r > x_\ell$ and every $\lambda \geq 0$, so $r = x_\ell$ is optimal. If $A^*(p) = 0$, an aspirational reference point can therefore tie only when $\lambda = 0$. If $A^*(p) < 0$, then at $\lambda = 0$, $r = x_\ell$ is the only optimal reference point.

Now suppose that $A^*(p) > 0$ and recall that

$$\bar{\lambda}(p) = \frac{\pi}{1-\pi}A^*(p).$$

If $\lambda < \bar{\lambda}(p)$, by the definition of the supremum there exists some $r > x_\ell$ for which the term in brackets is positive. Hence $F(r; p, \lambda) > F(x_\ell; p, \lambda)$, so every optimal reference point is aspirational.

If $\lambda > \bar{\lambda}(p)$, then for every $r > x_\ell$,

$$\frac{\pi}{1-\pi}\frac{u(r) + g(x_h - r) - u(x_\ell) - g(x_h - x_\ell)}{\ell(r - x_\ell)} - \lambda < 0.$$

Hence every aspirational reference point gives strictly lower utility than x_ℓ, so $r = x_\ell$ is the unique optimum.

Finally, if $\lambda = \bar{\lambda}(p)$, the same expression is weakly negative for every $r > x_\ell$. Thus x_ℓ is optimal, and an aspirational reference point can tie if the supremum defining $A^*(p)$ is attained at some $r > x_\ell$. ∎

A.3.3 Proof of Proposition 3

Proof. By Proposition 2, if $A^*(p) \leq 0$, or if $A^*(p) > 0$ and $\lambda \geq \bar{\lambda}(p)$, then $r = x_\ell$ is optimal. Hence $V(p; \lambda) = V^{NL}(p)$, and therefore $\bar{y}(p; \lambda) = \bar{y}^{NL}(p)$.

Now suppose $A^*(p) > 0$ and take $0 \leq \lambda_1 < \lambda_2 < \bar{\lambda}(p)$. Let $r_2 > x_\ell$ be optimal at λ_2. Since $\ell(r_2 - x_\ell) > 0$,

$$V(p;\lambda_2) = \mathbb{E}_p[U(X, r_2;\lambda_2)] < \mathbb{E}_p[U(X, r_2;\lambda_1)] \leq V(p;\lambda_1).$$

Thus $V(p;\lambda)$ is strictly decreasing for $\lambda < \bar{\lambda}(p)$. Since $S(y)$ is strictly increasing, $\bar{y}(p;\lambda)$ is also strictly decreasing on this region. ■

A.3.4 Proof of corollary 1

Proof. If $A^*(p) \leq 0$, or if $\bar{\lambda}(p) \leq \lambda_1 < \lambda_2$, Proposition 3 gives $\bar{y}(p;\lambda_1) = \bar{y}(p;\lambda_2) = \bar{y}^{NL}(p)$. Now suppose $\lambda_1 < \bar{\lambda}(p)$. By Proposition 2, the λ_1 type chooses an aspirational reference point, so $\bar{y}(p;\lambda_1) > \bar{y}^{NL}(p)$. If $\lambda_2 \geq \bar{\lambda}(p)$, then $\bar{y}(p;\lambda_2) = \bar{y}^{NL}(p)$. If instead $\lambda_2 < \bar{\lambda}(p)$, strict monotonicity from Proposition 3 gives $\bar{y}(p;\lambda_1) > \bar{y}(p;\lambda_2)$. ■

A.3.5 Proof of corollary 2

Proof. By definition of the certainty equivalent,

$$p \succeq_{M(y)} \delta_y \iff y \leq \bar{y}(p;\lambda).$$

Since $\bar{y}(p;\lambda_1) > \bar{y}(p;\lambda_2)$ whenever the two types differ only by their loss sensitivity, they make different choices if and only if $\bar{y}(p;\lambda_2) < y < \bar{y}(p;\lambda_1)$. ■

A.3.6 Proof of proposition 4

Proof. Since

$$\pi^A(\lambda) = \frac{\lambda}{\lambda + A^*(x_\ell, x_h)},$$

we have $\pi^A(\lambda_1) < \pi^A(\lambda_2)$ whenever $\lambda_1 < \lambda_2$.

- If $\pi \leq \pi^A(\lambda_1)$, both types choose $r = x_\ell$. By Proposition 3, $\bar{y}(\pi;\lambda_1) = \bar{y}(\pi;\lambda_2)$, so $\Delta_y(\pi;\lambda_1,\lambda_2) = 0$.
- Now suppose $\pi > \pi^A(\lambda_1)$. The lower loss-sensitive type is aspirational. If $\pi \leq \pi^A(\lambda_2)$, the higher loss-sensitive type chooses $r = x_\ell$, so $\bar{y}(\pi;\lambda_1) > \bar{y}^{NL}(\pi) = \bar{y}(\pi;\lambda_2)$. If instead $\pi > \pi^A(\lambda_2)$, both types are aspirational and Proposition 3 gives $\bar{y}(\pi;\lambda_1) > \bar{y}(\pi;\lambda_2)$. Hence $\Delta_y(\pi;\lambda_1,\lambda_2) > 0$ whenever $\pi > \pi^A(\lambda_1)$.

Finally, for any $r \in [x_\ell, x_h]$,

$$0 \leq \mathbb{E}_{p_\pi}[U(X, r;\lambda_1)] - \mathbb{E}_{p_\pi}[U(X, r;\lambda_2)] \leq (1-\pi)(\lambda_2 - \lambda_1)\ell(x_h - x_\ell).$$

Taking the maxima over r gives

$$0 \leq V(p_\pi; \lambda_1) - V(p_\pi; \lambda_2) \leq (1-\pi)(\lambda_2 - \lambda_1)\ell(x_h - x_\ell).$$

The right-hand side converges to zero as $\pi \to 1$. Since S is continuous and strictly increasing, then also $\Delta_y(\pi; \lambda_1, \lambda_2) \to 0$. ■

A.3.7 Proof of Lemma 3

Proof. When the optimal reference point is no-loss, $r_p^*(d, \lambda) = x_\ell(d) = \mu - \pi d$ which decreases with d. Now consider the case where the optimal reference point is aspirational. The first-order condition is

$$F(r, d) := \pi\big[u'(r) - g'(x_h(d) - r)\big] - (1-\pi)\lambda\ell'(r - x_\ell(d)) = 0.$$

At an interior local maximum, $F_r < 0$. Holding r fixed,

$$F_d = -\pi(1-\pi)g''(x_h(d) - r) - \pi(1-\pi)\lambda\ell''(r - x_\ell(d)).$$

Since $g'' < 0$ and $\ell'' < 0$, we have $F_d > 0$. By the implicit function theorem,

$$\frac{dr_p^*(d, \lambda)}{dd} = -\frac{F_d}{F_r} > 0.$$

■

A.3.8 Proof of Proposition 5.

Proof. For $d > 0$, and $t = r - x_\ell \in [0, d]$ rewrite the aspirational index as

$$a(t, d) := \frac{u(\mu - \pi d + t) + g(d - t) - u(\mu - \pi d) - g(d)}{\ell(t)}$$

For fixed $t > 0$, the derivative of the numerator with respect to d is

$$\pi\big[u'(\mu - \pi d) - u'(\mu - \pi d + t)\big] + \big[g'(d - t) - g'(d)\big] > 0,$$

by strict concavity of u and g. Hence $a(t, d)$ is strictly increasing in d for every fixed t.

Take $d_2 > d_1$ and let t_1 maximize $a(t, d_1)$, using the continuous extension at $t = 0$. Since t_1 is also feasible at d_2,

$$A^*(d_2) \geq a(t_1, d_2) > a(t_1, d_1) = A^*(d_1).$$

Thus $A^*(d)$ is strictly increasing, and so is $\bar{\lambda}(d) = \frac{\pi}{1-\pi}A^*(d)$.

The statements about $\bar{d}(\lambda)$ then follow from Proposition 2.

Finally, suppose $g = u$. Let $x_\ell(d) = \mu - \pi d$. For $t \in (0, d]$, the numerator of $A^*(d)$ is

$$u(x_\ell(d) + t) - u(x_\ell(d)) - \big[u(d) - u(d - t)\big].$$

If $x_\ell(d) \geq d$, concavity of u implies that this expression is nonpositive for every t, so $A^*(d) \leq 0$. If $x_\ell(d) < d$, its derivative at $t = 0$ is

$$u'(x_\ell(d)) - u'(d) > 0,$$

so $A^*(d) > 0$. Hence $A^*(d) \le 0$ if and only if $x_\ell(d) \ge d$, or equivalently

$$d \le \frac{\mu}{1+\pi}.$$

■

A.3.9 Proof of corollary 3

Proof. If $d \le \bar{d}(\lambda_1)$, both types choose $r = x_\ell(d)$, so Proposition 3 gives $\bar{y}(d;\lambda_1) = \bar{y}(d;\lambda_2)$. Hence $\Delta_y(d;\lambda_1,\lambda_2) = 0$.

If $d > \bar{d}(\lambda_1)$, the less loss-sensitive type is aspirational. Proposition 3 then implies $\bar{y}(d;\lambda_1) > \bar{y}(d;\lambda_2)$. The high loss-sensitive type can either choose an aspirational or no-loss reference point. Hence $\Delta_y(d;\lambda_1,\lambda_2) > 0$. ■

A.4 Comparison with expectations-based reference points

Consider a binary lottery $p = (x_h, \pi; x_\ell, 1-\pi)$ with $x_h > x_\ell$ and $\pi \in (0,1)$, and write $D = x_h - x_\ell$.

Kőszegi–Rabin. Under choice-acclimating personal equilibrium (Kőszegi and Rabin, 2007), each lottery is evaluated against its own outcome distribution. An outcome x evaluated against a reference outcome r gives utility

$$m(x) + \mu\big(m(x) - m(r)\big), \qquad \mu(z) = \begin{cases} \eta z, & z \ge 0, \\ \eta\lambda^{KR} z, & z < 0, \end{cases}$$

where m is increasing consumption utility, $\eta > 0$ is the weight on gain–loss utility, and $\lambda^{KR} \ge 1$ is the coefficient of loss aversion. Let $\Delta m = m(x_h) - m(x_\ell) > 0$. With probability $\pi(1-\pi)$, the outcome is high and the reference outcome is low, which gives a gain $\eta\Delta m$. With the same probability, the outcome is low and the reference outcome is high, which gives a loss $-\eta\lambda^{KR}\Delta m$. All other comparisons give zero. Hence

$$V^{KR}(p;\lambda^{KR}) = \mathbb{E}_p[m(X)] - \eta(\lambda^{KR}-1)\pi(1-\pi)\Delta m,$$

and

$$\frac{\partial V^{KR}(p;\lambda^{KR})}{\partial \lambda^{KR}} = -\eta\pi(1-\pi)\Delta m < 0.$$

The value of every nondegenerate binary lottery therefore strictly decreases with loss aversion. A sure payoff y is evaluated against itself and has value $m(y)$, which does not depend on λ^{KR}. The certainty equivalent of p, when it exists, therefore also strictly decreases with λ^{KR}.

Same utility, lottery-based reference point. Now keep the utility function $U(x,r)$ of Section 2, with loss sensitivity λ, but let the reference point be drawn from the lottery itself instead of being chosen. With X and R independent draws from p, the value of the lottery is

$$\widetilde{V}(p;\lambda) = \mathbb{E}\big[U(X,R)\big].$$

The outcome falls below the reference point only when $X = x_\ell$ and $R = x_h$, which happens with probability $\pi(1-\pi)$. Utility is then $u(x_\ell) - \lambda\ell(D)$. In the other three cases, λ plays no role. Hence

$$\frac{\partial \widetilde{V}(p;\lambda)}{\partial \lambda} = -\pi(1-\pi)\ell(D) < 0.$$

Even with the same utility function, the value of a nondegenerate binary lottery strictly decreases with loss sensitivity when the reference point is drawn from the lottery.

In my model, the agent instead chooses a single reference point and can set $r = x_\ell$, which is attained in both states. When this choice is optimal, no outcome falls below the reference point, and the value of the lottery does not depend on λ.

A.5 Identification

A.5.1 Proof of Proposition 6

Proof. Consider first the lottery p_1 with dispersion $D_1 := x_{h1} - x_{\ell 1}$. By strict concavity of v_α, the function $H(r) = v_\alpha(r) + v_\alpha(x_{h1} - r)$ is maximized at $\frac{x_{h1}}{2}$. Since p_1 satisfies $\frac{x_{h1}}{2} \leq x_{\ell 1}$, then $H(r)$ is strictly decreasing on $(x_{\ell 1}, x_{h1}]$ and so $H(r) \leq H(x_{\ell 1})$.

For any $r > x_{\ell 1}$, expected utility of p_1 is

$$\pi_1 H(r) + (1-\pi_1)\left[v_\alpha(x_{\ell 1}) - \lambda v_\alpha(r - x_{\ell 1})\right].$$

The first term is no greater than at $r = x_{\ell 1}$ while the second term is non increasing in r for $\lambda \geq 0$. Hence the first lottery is evaluated at $r = x_{\ell 1}$ for every $\alpha < 1$ and $\lambda \geq 0$, giving

$$V_1(\alpha, \lambda) = v_\alpha(x_{\ell 1}) + \pi_1 v_\alpha(D_1).$$

The certainty equivalent satisfies $\bar{y}_1 \in (x_{\ell 1}, x_{h1})$. Indeed, the sure payoff $x_{\ell 1}$ has value $S_1(x_{\ell 1}; \alpha) = v_\alpha(x_{\ell 1}) < V_1(\alpha, \lambda)$, while the sure payoff x_{h1} evaluated at the reference point $r = x_{\ell 1}$ has value $S_1(x_{h1}; \alpha) \geq v_\alpha(x_{\ell 1}) + v_\alpha(D_1) > V_1(\alpha, \lambda)$ since $0 < \pi_1 < 1$. Furthermore, for any sure payoff $y \in [x_{\ell 1}, x_{h1}]$, we have

$$\frac{y}{2} \leq \frac{x_{h1}}{2} \leq x_{\ell 1}.$$

So by strict concavity of v_α, the sure payoff at the certainty equivalent $\bar{y}_1$ is optimally evaluated at $r = x_{\ell 1}$. Hence $S_1(\bar{y}_1; \alpha) = v_\alpha(x_{\ell 1}) + v_\alpha(\bar{y}_1 - x_{\ell 1})$. Since the lottery is indifferent to its certainty equivalent

$$\frac{v_\alpha(\bar{y}_1 - x_{\ell 1})}{v_\alpha(D_1)} = \pi_1$$

which is (5). To show that α is uniquely identified, I show that the ratio on the left-hand side must be strictly decreasing in α. This is equivalent to showing that for $\alpha_1 < \alpha_2 < 1$

$$\frac{v_{\alpha_1}(\bar{y}_1 - x_{\ell 1})}{v_{\alpha_1}(D_1)} > \frac{v_{\alpha_2}(\bar{y}_1 - x_{\ell 1})}{v_{\alpha_2}(D_1)}.$$

Since $\alpha_1 < \alpha_2 < 1$, v_{α_2} is a strictly convex transformation of v_{α_1}.

$$v_{\alpha_2}(z) = f(v_{\alpha_1}(z))$$

Indeed,

$$f'(v_{\alpha_1}(z)) = \frac{v'_{\alpha_2}(z)}{v'_{\alpha_1}(z)} = \frac{(z+\sigma)^{\alpha_2-1}}{(z+\sigma)^{\alpha_1-1}} = (z+\sigma)^{\alpha_2-\alpha_1} > 0.$$

Since $\alpha_2 - \alpha_1 > 0$, $f'(v_{\alpha_1}(z))$ is strictly increasing in z, so f is strictly convex. By $\bar{y}_1 \in (x_{\ell 1}, x_{h1})$, we have $\bar{y}_1 - x_{\ell 1} < D_1$, and since $f(0) = 0$, strict convexity implies

$$\frac{v_{\alpha_1}(\bar{y}_1 - x_{\ell 1})}{v_{\alpha_1}(D_1)} > \frac{f(v_{\alpha_1}(\bar{y}_1 - x_{\ell 1}))}{f(v_{\alpha_1}(D_1))} = \frac{v_{\alpha_2}(\bar{y}_1 - x_{\ell 1})}{v_{\alpha_2}(D_1)}.$$

Hence α is uniquely identified.

Now fix $\alpha = \widehat{\alpha}$. Since the second certainty equivalent is strictly above the no-loss benchmark and S_2 is strictly increasing, we have

$$V_2(\widehat{\alpha}, \lambda) = S_2(\bar{y}_2; \widehat{\alpha}) > S_2\big(\bar{y}_2^{NL}(\widehat{\alpha}); \widehat{\alpha}\big) = V_2^{NL}(\widehat{\alpha}).$$

A higher λ strictly lowers V_2 whenever the optimum is aspirational. If $r_b > x_{\ell 2}$ is optimal under λ_b and $\lambda_a < \lambda_b$, then

$$V_2(\widehat{\alpha}, \lambda_a) \geq V_2(\widehat{\alpha}, \lambda_b) + (\lambda_b - \lambda_a)(1 - \pi_2) v_{\widehat{\alpha}}(r_b - x_{\ell 2}) > V_2(\widehat{\alpha}, \lambda_b).$$

Since the certainty equivalent lies above the no-loss benchmark, the optimum is aspirational. Moreover V_2 is strictly decreasing in λ when the optimum is aspirational so λ is uniquely identified by $V_2(\widehat{\alpha}, \lambda) = S_2(\bar{y}_2; \widehat{\alpha})$. Therefore α and λ are uniquely identified and (α, λ) is globally point identified.

■

A.6 Reference-point adjustment

Lemma A.3 (Loss sensitivity and reference-point adjustment)**.** *Consider a stable interior aspirational optimum (a^*, r^*) of $F(a, r; \lambda)$, with $F_{a\lambda} > 0$ and $F_{r\lambda} < 0$.*

- *If $F_{ar} < 0$, then*

$$\frac{dr^*}{d\lambda} < 0, \qquad \frac{da^*}{d\lambda} > \left.\frac{\partial a^{\text{fix}}}{\partial \lambda}\right|_{r=r^*} > 0.$$

- *If $F_{ar} > 0$ and $dr^*/d\lambda < 0$, then*

$$\frac{da^*}{d\lambda} < \left.\frac{\partial a^{\text{fix}}}{\partial \lambda}\right|_{r=r^*}.$$

Thus a lower reference point reinforces the action response when $F_{ar} < 0$ and offsets it when $F_{ar} > 0$. In the second case, the total action response need not be positive.

A.6.1 Proof of Lemma A.3

Proof. Denote the determinant of the Hessian of F with respect to (a, r) by $D := F_{aa}F_{rr} - F_{ar}^2$. Since the aspirational optimum is stable, the Hessian is negative definite which implies $F_{aa} < 0$ and $D > 0$.

Differentiating the two first-order conditions with respect to λ gives

$$\frac{dr^*}{d\lambda} = \frac{F_{ar}F_{a\lambda} - F_{aa}F_{r\lambda}}{D}.$$

If $F_{ar} < 0$, $F_{a\lambda} > 0$, $F_{r\lambda} < 0$, and $F_{aa} < 0$ then $\frac{dr^*}{d\lambda} < 0$. Since also

$$\left.\frac{\partial a^{\text{fix}}}{\partial \lambda}\right|_{r=r^*} = -\frac{F_{a\lambda}}{F_{aa}} > 0,$$

equation (6) then gives

$$\frac{da^*}{d\lambda} > \left.\frac{\partial a^{\text{fix}}}{\partial \lambda}\right|_{r=r^*}.$$

On the other hand, if $F_{ar} > 0$ and $dr^*/d\lambda < 0$, equation (6) gives

$$\frac{da^*}{d\lambda} < \left.\frac{\partial a^{\text{fix}}}{\partial \lambda}\right|_{r=r^*}.$$

■

A.7 Proofs for the portfolio application

A.7.1 Proof of Proposition 7

Proof. **No-loss choice.** Using the budget constraint, rewrite $x_H = m + e^{-\rho}(m - x_L)$. So, at an interior no-loss optimum, $0 < r < x_L < m$, the objective can be written as a function of x_L and r

$$F^{NL}(x_L, r; \rho) = u(r) + \frac{1}{2}g(x_H - r) + \frac{1}{2}g(x_L - r).$$

The two first-order conditions for the portfolio and the reference point are

$$F^{NL}_{x_L} = \frac{1}{2}\left[g'(x_L - r) - e^{-\rho}g'(x_H - r)\right] = 0,$$
$$F^{NL}_{r} = u'(r) - \frac{1}{2}\left[g'(x_H - r) + g'(x_L - r)\right] = 0.$$

The second derivatives and cross partials are

$$F^{NL}_{x_Lx_L} = \frac{1}{2}\left[e^{-2\rho}g''(x_H - r) + g''(x_L - r)\right] < 0,$$
$$F^{NL}_{rr} = u''(r) + \frac{1}{2}\left[g''(x_H - r) + g''(x_L - r)\right] < 0,$$
$$F^{NL}_{x_Lr} = \frac{1}{2}\left[e^{-\rho}g''(x_H - r) - g''(x_L - r)\right].$$

Denote by D the determinant of the Hessian of the no loss objective F^{NL} with respect to (x_L, r): $D = F^{NL}_{x_Lx_L}F^{NL}_{rr} - \left(F^{NL}_{x_Lr}\right)^2$.

$$D = \frac{1}{2}u''(r)\left[(e^{-2\rho})g''(x_H - r) + g''(x_L - r)\right] + \frac{1}{4}(1 + e^{-\rho})^2 g''(x_H - r)g''(x_L - r) > 0.$$

Hence the Hessian is negative definite. The partial derivatives with respect to prices, holding (x_L, r) fixed, are

$$F^{NL}_{x_L\rho} = \frac{1}{2}e^{-\rho}\left[g'(x_H - r) + (x_H - m)g''(x_H - r)\right],$$
$$F^{NL}_{r\rho} = \frac{1}{2}(x_H - m)g''(x_H - r) < 0.$$

Differentiate both first-order conditions with respect to ρ:

$$F^{NL}_{x_Lx_L}\frac{dx^{NL}_L}{d\rho} + F^{NL}_{x_Lr}\frac{dr^{NL}}{d\rho} + F^{NL}_{x_L\rho} = 0,$$
$$F^{NL}_{rr}\frac{dr^{NL}}{d\rho} + F^{NL}_{rx_L}\frac{dx^{NL}_L}{d\rho} + F^{NL}_{r\rho} = 0.$$

So rearranging the first equation gives

$$\frac{dx^{NL}_L}{d\rho} = -\frac{F^{NL}_{x_L\rho}}{F^{NL}_{x_Lx_L}} - \frac{F^{NL}_{x_Lr}}{F^{NL}_{x_Lx_L}}\frac{dr^{NL}}{d\rho}.$$

And substituting it in the second equation gives

$$\frac{dr^{NL}}{d\rho} = \frac{F^{NL}_{rx_L}F^{NL}_{x_L\rho} - F^{NL}_{r\rho}F^{NL}_{x_Lx_L}}{F^{NL}_{rr}F^{NL}_{x_Lx_L} - \left(F^{NL}_{rx_L}\right)^2}.$$

The denominator is the determinant of the Hessian which is positive.

Sign of $\frac{dr^{NL}}{d\rho}$. I first look at the effect of the prices on the optimal reference point.

$$\begin{aligned}4D\frac{dr^{NL}}{d\rho} =& e^{-\rho}\left[e^{-\rho}g''(x_H - r) - g''(x_L - r)\right]\left[g'(x_H - r) + (x_H - m)g''(x_H - r)\right]\\ &- \left[e^{-2\rho}g''(x_H - r) + g''(x_L - r)\right](x_H - m)g''(x_H - r)\end{aligned}$$

Cancelling terms and using the portfolio FOC $e^{-\rho}g'(x_H - r) = g'(x_L - r)$ gives

$$\begin{aligned}4D\frac{dr^{NL}}{d\rho} &= \left[e^{-\rho}g''(x_H - r) - g''(x_L - r)\right]g'(x_L - r)\\ &\quad - (1 + e^{-\rho})(x_H - m)g''(x_H - r)g''(x_L - r)\\ \iff \frac{4e^{\rho}D}{g''(x_L - r)g''(x_H - r)}\frac{dr^{NL}}{d\rho} &= \frac{g'(x_H - r)}{-g''(x_H - r)} - \frac{g'(x_L - r)}{-g''(x_L - r)} - (x_H - m)(1 + e^{\rho})\end{aligned}$$

The budget line gives $x_H - x_L = (x_H - m)(1 + e^{\rho})$ so

$$\frac{4e^{\rho}D}{g''(x_L - r)g''(x_H - r)}\frac{dr^{NL}}{d\rho} = \frac{g'(x_H - r)}{-g''(x_H - r)} - \frac{g'(x_L - r)}{-g''(x_L - r)} - (x_H - x_L)$$

Since the term multiplying $\frac{dr^{NL}}{d\rho}$ is positive, then

$$\frac{dr^*}{d\rho} > 0 \quad \iff \quad \frac{g'(x_H - r)}{-g''(x_H - r)} - \frac{g'(x_L - r)}{-g''(x_L - r)} > (x_H - x_L)$$

Now suppose g has the shifted-power form with $0 < \alpha_g < 1$. Then

$$g'(x) = (x + \sigma)^{\alpha_g - 1},$$
$$g''(x) = (\alpha_g - 1)(x + \sigma)^{\alpha_g - 2}.$$

Substituting into the condition above gives

$$\frac{dr^*}{d\rho} > 0 \quad \iff \quad \frac{\alpha_g}{1 - \alpha_g}(x_H - x_L) > 0 \tag{17}$$

Hence $dr^{NL}/d\rho > 0$ for $0 < \alpha_g < 1$.

Sign of $\frac{dx_L^{NL}}{d\rho}$. I now derive the sign of the effect of prices on the optimal demand. For the remaining comparisons, suppose $\alpha_g \in (0, 1)$. Recall that the expression is

$$\frac{dx_L^{NL}}{d\rho} = -\frac{F_{x_L\rho}^{NL}}{F_{x_Lx_L}^{NL}} - \frac{F_{x_Lr}^{NL}}{F_{x_Lx_L}^{NL}} \frac{dr^{NL}}{d\rho}.$$

The only signs left to be determined are $F_{x_Lr}^{NL}$ and $F_{x_L\rho}^{NL}$.

$$\begin{aligned} F_{x_Lr}^{NL} &= \frac{1}{2}\left[e^{-\rho} g''(x_H - r) - g''(x_L - r)\right] \\ &= \frac{1}{2}\left[\frac{g'(x_L - r)}{g'(x_H - r)} g''(x_H - r) - g''(x_L - r)\right] \\ &= \frac{g''(x_H - r)g''(x_L - r)}{2g'(x_H - r)}\left[\frac{g'(x_L - r)}{g''(x_L - r)} - \frac{g'(x_H - r)}{g''(x_H - r)}\right] > 0. \end{aligned}$$

By strict concavity of g, the first term is positive, and by (17), whenever $\frac{dr^{NL}}{d\rho} > 0$, then the term in parentheses is positive and $F_{x_Lr}^{NL} > 0$.

$$\begin{aligned} F_{x_L\rho}^{NL} &= \frac{1}{2} e^{-\rho}\left[g'(x_H - r) + (x_H - m)g''(x_H - r)\right] \\ &= \frac{e^{-\rho}[-g''(x_H - r)]}{2}\left[-\frac{g'(x_H - r)}{g''(x_H - r)} - (x_H - m)\right]. \end{aligned}$$

$$\text{By} \quad \frac{g'(x_H - r)}{-g''(x_H - r)} > \frac{g'(x_H - r)}{-g''(x_H - r)} - \frac{g'(x_L - r)}{-g''(x_L - r)} > (x_H - x_L) \geq x_H - m$$

the second term is positive. So $F_{x_L\rho}^{NL} > 0$.

Hence

$$\frac{dx_L^{NL}}{d\rho} = -\frac{F_{x_L\rho}^{NL}}{F_{x_Lx_L}^{NL}} - \frac{F_{x_Lr}^{NL}}{F_{x_Lx_L}^{NL}} \frac{dr^{NL}}{d\rho}.$$

Then $\frac{dx_L^{NL}}{d\rho} > 0$ whenever $\frac{dr^{NL}}{d\rho} > 0$, which happens when $\alpha_g \in (0, 1)$.

Here $x_L(\rho, r)$ denotes demand conditional on r, and both responses are evaluated at the same initial

allocation and reference point. By the implicit function theorem, we can rewrite

$$\frac{dx_L^{NL}}{d\rho} - \left.\frac{\partial x_L(\rho, r)}{\partial \rho}\right|_{r=r^{NL}} = -\frac{F_{x_L r}^{NL}}{F_{x_L x_L}^{NL}} \frac{dr^{NL}}{d\rho}$$

Which is greater than 0 for $\alpha_g \in (0, 1)$.

Sign of relative demand. The budget gives $x_H = m + e^{-\rho}(m - x_L(\rho))$. Then

$$\frac{dx_H}{d\rho} = -(x_H - m) - e^{-\rho}\frac{dx_L}{d\rho} < 0$$

Now looking at the derivative of the relative demand and substituting the expression just above gives

$$\begin{aligned}\frac{ds(\rho)}{d\rho} &= \frac{x_L \frac{dx_H}{d\rho} - x_H \frac{dx_L}{d\rho}}{(x_L + x_H)^2} \\ &= \frac{-x_L(x_H - m) - (x_H + x_L e^{-\rho})\frac{dx_L}{d\rho}}{(x_L + x_H)^2} < 0\end{aligned}$$

Comparing with a fixed reference point:

$$\frac{ds(\rho)}{d\rho} - \left.\frac{\partial s(\rho, r)}{\partial \rho}\right|_{r=r^{NL}} = \frac{-(x_H + x_L e^{-\rho})}{(x_L + x_H)^2}\left[\frac{dx_L(\rho)}{d\rho} - \left.\frac{\partial x_L(\rho, r)}{\partial \rho}\right|_{r=r^{NL}}\right]$$

where the expression in brackets is positive and the coefficient is negative. Hence

$$\frac{ds(\rho)}{d\rho} - \left.\frac{\partial s(\rho, r)}{\partial \rho}\right|_{r=r^{NL}} < 0$$

Since $\left.\frac{\partial x_L(\rho,r)}{\partial \rho}\right|_{r=r^{NL}} = -\frac{F_{x_L\rho}^{NL}}{F_{x_L x_L}^{NL}} > 0$, the same relative-demand formula gives $\left.\frac{\partial s(\rho,r)}{\partial \rho}\right|_{r=r^{NL}} < 0$.

Aspirational choice. Fix $\rho < 0$ and consider a stable interior aspirational optimum, with a negative-definite Hessian in (x_L, r). All derivatives are evaluated at the initial optimum, holding m and λ fixed. At an interior aspirational optimum, $0 < x_L < m$ and $x_L < r < x_H$, the objective is

$$F^A(x_L, r; \rho, \lambda) = \frac{1}{2}\big[u(r) + g(x_H - r) + u(x_L) - \lambda\ell(r - x_L)\big].$$

The budget gives $x_H = m + e^{-\rho}(m - x_L) > m$, and all derivatives below use this substitution.

The two first-order conditions are

$$\begin{aligned}F_{x_L}^A &= \frac{1}{2}\big[u'(x_L) + \lambda\ell'(r - x_L) - e^{-\rho}g'(x_H - r)\big] = 0, \\ F_r^A &= \frac{1}{2}\big[u'(r) - g'(x_H - r) - \lambda\ell'(r - x_L)\big] = 0.\end{aligned}$$

The second derivatives and cross partials are

$$F^A_{x_Lx_L} = \frac{1}{2}\left[e^{-2\rho}g''(x_H - r) + u''(x_L) - \lambda\ell''(r - x_L)\right],$$
$$F^A_{rr} = \frac{1}{2}\left[u''(r) + g''(x_H - r) - \lambda\ell''(r - x_L)\right],$$
$$F^A_{x_Lr} = \frac{1}{2}\left[e^{-\rho}g''(x_H - r) + \lambda\ell''(r - x_L)\right] < 0.$$

By assumption, the Hessian is negative definite, so $F^A_{rr}, F^A_{x_Lx_L} < 0$ and $D = F^A_{rr}F^A_{x_Lx_L} - (F^A_{x_Lr})^2 > 0$.

The price partials are

$$F^A_{x_L\rho} = \frac{1}{2}e^{-\rho}\left[g'(x_H - r) + (x_H - m)g''(x_H - r)\right],$$
$$F^A_{r\rho} = \frac{1}{2}(x_H - m)g''(x_H - r) < 0.$$

Using the general derivative expressions derived above,

$$\frac{dx^A_L}{d\rho} = -\frac{F^A_{x_L\rho}}{F^A_{x_Lx_L}} - \frac{F^A_{x_Lr}}{F^A_{x_Lx_L}}\frac{dr^A}{d\rho},$$
$$\frac{dr^A}{d\rho} = \frac{F^A_{rx_L}F^A_{x_L\rho} - F^A_{r\rho}F^A_{x_Lx_L}}{D}.$$

Sign of $\frac{dr^A}{d\rho}$. Since $F^A_{rx_L} < 0$, $F^A_{r\rho} < 0$, and $F^A_{x_Lx_L} < 0$, a sufficient condition for $dr^A/d\rho < 0$ is $F^A_{x_L\rho} \geq 0$. Indeed, the two terms in the numerator of $dr^A/d\rho$ are then weakly and strictly negative, respectively.

Holding the reference point fixed,

$$\left.\frac{\partial x_L(\rho, r)}{\partial\rho}\right|_{r=r^A} = -\frac{F^A_{x_L\rho}}{F^A_{x_Lx_L}},$$

so $F^A_{x_L\rho} \geq 0$ corresponds to the standard local price response of the matched fixed-reference investor.

Using

$$F^A_{x_L\rho} = \frac{e^{-\rho}[-g''(x_H - r)]}{2}\left[-\frac{g'(x_H - r)}{g''(x_H - r)} - (x_H - m)\right],$$

I therefore maintain condition (9) from the main text,

$$-\frac{g'(x_H - r)}{g''(x_H - r)} \geq x_H - m.$$

This condition is sufficient, not necessary, for the conclusions below.

Under the shifted-power specification, $-g'(x_H - r)/g''(x_H - r) = (x_H - r + \sigma)/(1 - \alpha_g)$. Hence, for $0 < \alpha_g < 1$, condition (9) is equivalent to

$$r \leq m + \sigma + \alpha_g(x_H - m) = (1 - \alpha_g)m + \alpha_g x_H + \sigma.$$

Thus the condition allows $r > m$. If $\sigma \geq (1 - \alpha_g)(x_H - m)$, it is automatically satisfied for every interior aspirational reference point $r < x_H$.

Under condition (9), $F^A_{x_L\rho} \geq 0$, and therefore $dr^A/d\rho < 0$.

Sign of $\frac{dx_L^*}{d\rho}$ and comparison with a fixed reference point. Using the derivative expression above, the first term in $dx_L^A/d\rho$ is nonnegative and the second is strictly positive since $F^A_{x_Lr} < 0$, $F^A_{x_Lx_L} < 0$, and $dr^A/d\rho < 0$. Hence $dx_L^A/d\rho > 0$, and therefore $d(r^A - x_L^A)/d\rho < 0$.

Here $x_L(\rho, r)$ denotes demand conditional on r, with both responses evaluated at the same initial allocation and reference point. Since

$$\left.\frac{\partial x_L(\rho, r)}{\partial \rho}\right|_{r=r^A} = -\frac{F^A_{x_L\rho}}{F^A_{x_Lx_L}} \geq 0,$$

Thus equation (6) gives

$$\frac{dx_L^A}{d\rho} - \left.\frac{\partial x_L(\rho, r)}{\partial \rho}\right|_{r=r^A} = -\frac{F^A_{x_Lr}}{F^A_{x_Lx_L}}\frac{dr^A}{d\rho} > 0.$$

Hence

$$\frac{dx_L^A}{d\rho} > \left.\frac{\partial x_L(\rho, r)}{\partial \rho}\right|_{r=r^A} \geq 0.$$

Relative demand. Using the relative-demand expression derived above,

$$\frac{ds}{d\rho} = -\frac{x_L(x_H - m) + m(1 + e^{-\rho})\frac{dx_L}{d\rho}}{(x_L + x_H)^2}.$$

Since $x_L > 0$, $x_H > m$, and both lower-payoff responses are nonnegative,

$$\frac{ds^A}{d\rho} < \left.\frac{\partial s(\rho, r)}{\partial \rho}\right|_{r=r^A} < 0,$$

where the first inequality follows from

$$\frac{ds^A}{d\rho} - \left.\frac{\partial s(\rho, r)}{\partial \rho}\right|_{r=r^A} = -\frac{m(1 + e^{-\rho})}{(x_L + x_H)^2}\left[\frac{dx_L^A}{d\rho} - \left.\frac{\partial x_L(\rho, r)}{\partial \rho}\right|_{r=r^A}\right] < 0.$$

Thus, under condition (9), a small fall in ρ raises relative demand for the cheaper state in both problems, and the increase is larger when the reference point can adjust. ■

A.7.2 Proof of Corollary 4

Proof. At $\rho = 0$, the budget is $x_H + x_L = 2m$. By symmetry, take $x_H \geq x_L$.

If $r \leq x_L$, concavity of g gives

$$u(r) + \frac{1}{2}g(x_H - r) + \frac{1}{2}g(x_L - r) \leq u(r) + g(m - r).$$

Thus, for any no-loss reference point, the symmetric portfolio (m, m) gives a weakly higher value.

If $x_L < r \leq x_H$, the loss term is nonpositive, so

$$\begin{aligned}
&\frac{1}{2}\big[u(r) + g(x_H - r)\big] + \frac{1}{2}\big[u(x_L) - \lambda\ell(r - x_L)\big] \\
&\leq \frac{1}{2}\big[u(r) + g(x_H - r)\big] + \frac{1}{2}u(x_L) \\
&\leq u\left(\frac{r + x_L}{2}\right) + g\left(\frac{x_H - r}{2}\right) \\
&= u\left(\frac{r + x_L}{2}\right) + g\left(m - \frac{r + x_L}{2}\right).
\end{aligned}$$

where the inequality follows from concavity of u and g and $g(0) = 0$. The last expression is the value of the symmetric portfolio with reference point $(r + x_L)/2 \leq m$. This reference point is no-loss. Moreover, since $r > x_L$, the inequality is strict. Thus an aspirational allocation cannot be optimal at $\rho = 0$.

Thus the optimal portfolio at $\rho = 0$ is symmetric and the optimal reference point is no-loss. Hence $s(0, m, \lambda) = \frac{1}{2}$ for every $\lambda \geq 0$.

Let $r_m < m$ denote the unique no-loss optimum at $\rho = 0$. Since $u(r_m) + g(m - r_m) > u(m)$, the no-loss value strictly exceeds $u(m)$. To establish a strict gap also at the boundary of the aspirational region, set $t = (r + x_L)/2 \in [0, m]$. The inequality above gives, for every aspirational allocation at $\lambda = 0$,

$$F^A(x_L, r; 0, 0) \leq u(t) + \frac{1}{2}g\big(2(m - t)\big).$$

For $t < m$, strict concavity of g and $g(0) = 0$ imply

$$u(t) + \frac{1}{2}g\big(2(m - t)\big) < u(t) + g(m - t) \leq V^{NL}(0, m).$$

For $t = m$, this expression is $u(m) < V^{NL}(0, m)$. Since the expression is continuous on the compact interval $[0, m]$, its maximum is strictly below $V^{NL}(0, m)$. Hence

$$V^A(0, m, 0) < V^{NL}(0, m).$$

By continuity, there is some $\varepsilon > 0$ such that

$$V^A(\rho, m, 0) < V^{NL}(\rho, m)$$

for every $|\rho| < \varepsilon$.

For $\lambda \geq 0$, the no-loss value does not change, while $V^A(\rho, m, \lambda) \leq V^A(\rho, m, 0)$. The same neighborhood is therefore no-loss for every λ. Since λ does not enter F^{NL}, the optimal portfolio and reference point are independent of λ when no-loss. ∎

A.7.3 Proof of Proposition 8

Proof. Fix $\rho \leq 0$, so the cheapest asset satisfies $x_H \geq x_L$. Since no reference points above x_H are optimal, the expected utility can be written as

$$F(x_H, x_L, r; \lambda) = F(x_H, x_L, r; 0) - \frac{\lambda}{2}\ell\big((r - x_L)_+\big).$$

Take any optimizer (x_H^1, x_L^1, r^1) at λ_1 and any optimizer (x_H^2, x_L^2, r^2) at λ_2 such that $\lambda_2 > \lambda_1$. Optimality

gives

$$F(x_H^1, x_L^1, r^1; 0) - \frac{\lambda_1}{2}\ell\big((r^1 - x_L^1)_+\big) \geq F(x_H^2, x_L^2, r^2; 0) - \frac{\lambda_1}{2}\ell\big((r^2 - x_L^2)_+\big),$$
$$F(x_H^2, x_L^2, r^2; 0) - \frac{\lambda_2}{2}\ell\big((r^2 - x_L^2)_+\big) \geq F(x_H^1, x_L^1, r^1; 0) - \frac{\lambda_2}{2}\ell\big((r^1 - x_L^1)_+\big).$$

Adding the two inequalities gives

$$(\lambda_2 - \lambda_1)\left[\ell\big((r^1 - x_L^1)_+\big) - \ell\big((r^2 - x_L^2)_+\big)\right] \geq 0.$$

Since ℓ is strictly increasing and $\lambda_2 - \lambda_1 > 0$,

$$(r^2 - x_L^2)_+ \leq (r^1 - x_L^1)_+.$$

Thus at any $\rho \leq 0$, more loss-sensitive investor chooses a weakly smaller shortfall. Therefore, if the investor with λ_1 chooses zero shortfall, every optimizer for the investor with $\lambda_2 > \lambda_1$ also has zero shortfall.

I now show that the set of prices for which a more loss-sensitive investor aspires is contained in the set of prices for which a less loss-sensitive investor aspires. Here if $\rho \in \mathcal{A}(\lambda_2, m)$,

$$V^A(\rho, m, \lambda_2) > V^{NL}(\rho, m),$$

Furthermore the no-loss value $V^{NL}(\rho, m)$ is independent of λ, while $V^A(\rho, m, \lambda)$ is weakly decreasing in λ which implies

$$V^A(\rho, m, \lambda_1) \geq V^A(\rho, m, \lambda_2) > V^{NL}(\rho, m).$$

Thus $\rho \in \mathcal{A}(\lambda_1, m)$ and $\mathcal{A}(\lambda_2, m) \subseteq \mathcal{A}(\lambda_1, m)$. The case $\rho > 0$ is symmetric. ■

A.7.4 Proof of Proposition 9

Proof. The no-loss objective does not depend on λ, so the portfolio and reference point do not vary with λ as long as the no-loss regime is optimal. Now consider a stable interior aspirational optimum. The cross partials are

$$F^A_{x_L\lambda} = \frac{1}{2}\ell'(r - x_L) > 0,$$
$$F^A_{r\lambda} = -\frac{1}{2}\ell'(r - x_L) < 0,$$
$$F^A_{x_L r} = \frac{1}{2}\left[e^{-\rho}g''(x_H - r) + \lambda\ell''(r - x_L)\right] < 0.$$

Lemma A.3 therefore gives

$$\frac{dr^A}{d\lambda} < 0, \qquad \frac{dx_L^A}{d\lambda} > \left.\frac{\partial x_L^{\text{fix}}}{\partial\lambda}\right|_{r=r^A} > 0. \tag{18}$$

The budget constraint gives $x_H = m + e^{-\rho}(m - x_L)$ so $dx_H^A/d\lambda < 0$.

Hence the effects of loss sensitivity on payoff spread and shortfall are

$$\frac{d}{d\lambda}(x_H^A - x_L^A) < 0,$$
$$\frac{d}{d\lambda}(r^A - x_L^A) < 0.$$

Consider the relative demand $s = \frac{x_H}{x_H + x_L}$. The budget constraint implies

$$\frac{dx_H^A}{d\lambda} = -e^{-\rho}\frac{dx_L^A}{d\lambda},$$

so we have

$$\frac{ds^A}{d\lambda} = \frac{x_L\, dx_H^A/d\lambda - x_H\, dx_L^A/d\lambda}{(x_H + x_L)^2} = -\frac{x_H + e^{-\rho}x_L}{(x_H + x_L)^2}\frac{dx_L^A}{d\lambda}.$$

Using the budget constraint again $x_H + e^{-\rho}x_L = m(1 + e^{-\rho})$ gives

$$\frac{ds^A}{d\lambda} = -\frac{m(1 + e^{-\rho})}{(x_H + x_L)^2}\frac{dx_L^A}{d\lambda}.$$

The same derivations can be done for the matched fixed-reference investor and

$$\left.\frac{\partial s^{\text{fix}}}{\partial \lambda}\right|_{r=r^A} = -\frac{m(1 + e^{-\rho})}{(x_H + x_L)^2}\left.\frac{\partial x_L^{\text{fix}}}{\partial \lambda}\right|_{r=r^A}.$$

By (18), both relative-demand responses are negative and

$$\frac{ds^A}{d\lambda} < \left.\frac{\partial s^{\text{fix}}}{\partial \lambda}\right|_{r=r^A} < 0.$$

■

A.8 Proofs for the effort application

A.8.1 General effort response

I define the shortfall and probability of success by $t := r - x_\ell$ and $p := qs(e)$. At an interior aspirational optimum, the cross partials are $F_{e\lambda} = qs'(e)\ell(t) > 0$ and $F_{r\lambda} = -(1 - p)\ell'(t) < 0$. Moreover, $F_{er} = qs'(e)\big[H'(r) - L_r(r, \lambda)\big]$. The reference-point first-order condition is

$$pH'(r) = (1 - p)\lambda\ell'(t),$$

so $H'(r) > 0$. Since $L_r(r, \lambda) = -\lambda\ell'(t) < 0$, it follows that

$$F_{er} > 0.$$

Now consider a stable interior aspirational optimum and let $D := F_{ee}F_{rr} - F_{er}^2 > 0$ denote the determinant of the Hessian. Differentiating the two first-order conditions with respect to λ gives

$$\frac{de^*}{d\lambda} = \frac{-F_{e\lambda}F_{rr} + F_{er}F_{r\lambda}}{D}.$$

Using the reference-point first-order condition,

$$H'(r) - L_r(r, \lambda) = H'(r) + \lambda\ell'(t) = \frac{\lambda}{p}\ell'(t).$$

Substituting this expression in F_{er}, and using the cross partials above, the formula for $de^*/d\lambda$ rewrites as

$$\frac{de^*}{d\lambda} = \frac{qs'(e)}{D}\left[-F_{rr}\ell(t) - \frac{1-p}{p}\lambda\left[\ell'(t)\right]^2\right]. \tag{19}$$

This gives the two terms in (14).

To derive the sign of $de^*/d\lambda$, I rewrite the two terms in the bracket using the reference-point first-order condition.

First, consider F_{rr}. Using the first-order condition to substitute $\lambda(1-p)$

$$\begin{aligned} F_{rr} &= pH''(r) - (1-p)\lambda\ell''(t) \\ &= pH''(r) - pH'(r)\frac{\ell''(t)}{\ell'(t)} && \text{(by the FOC for } r\text{)} \\ &= -pH'(r)\left[-\frac{H''(r)}{H'(r)} + \frac{\ell''(t)}{\ell'(t)}\right]. \end{aligned}$$

Define the term into bracket by $\Delta(t) := -\frac{H''(x_\ell+t)}{H'(x_\ell+t)} + \frac{\ell''(t)}{\ell'(t)}$. Then

$$F_{rr} = -pH'(r)\Delta(t).$$

Next I consider the second term into bracket. Using the reference-point first-order condition again

$$\frac{1-p}{p}\lambda[\ell'(t)]^2 = H'(r)\ell'(t).$$

Substituting both expressions into (19) gives

$$\begin{aligned} \frac{de^*}{d\lambda} &= \frac{qs'(e)}{D}\left[pH'(r)\Delta(t)\ell(t) - H'(r)\ell'(t)\right] \\ &= \frac{qs'(e)H'(r)\ell'(t)}{D}\left[p\Delta(t)\frac{\ell(t)}{\ell'(t)} - 1\right]. \end{aligned}$$

The reference-point first-order condition also implies

$$p = \Pi(t;\lambda) := \frac{\lambda\ell'(t)}{H'(x_\ell + t) + \lambda\ell'(t)}.$$

Define $R(t,\lambda) := \Pi(t;\lambda)\Delta(t)\frac{\ell(t)}{\ell'(t)}$. Since the terms outside the bracket are positive,

$$\operatorname{sgn}\left(\frac{de^*}{d\lambda}\right) = \operatorname{sgn}\left[R(t^*,\lambda) - 1\right].$$

A.8.2 Proof of Theorem 2

Proof. Fix $\lambda > 0$. Under the parametric specification, $x_\ell = 0$. Define $u := u_{\alpha,\sigma}$. The shifted-power function satisfies $u(0) = 0$, $u'(z) = (z+\sigma)^{\alpha-1} > 0$, $u''(z) = (\alpha-1)(z+\sigma)^{\alpha-2} < 0$. The effort technology and cost function also satisfy the assumptions: $s(0) = 0$, $s'(e) = \beta e^{-\beta e} > 0$, $s''(e) = -\beta^2 e^{-\beta e} < 0$, while $c'(e) = \kappa > 0$ and $c''(e) = 0$.

Since $\phi(e) = \frac{\kappa}{\beta}e^{\beta e}$, we have $\phi'(e) = \kappa e^{\beta e} > 0$ and $\phi(e) \to \infty$ as $e \to \infty$, and conditional effort is finite.

Success and failure utilities are

$$H(r) = u(r) + u(x_h - r),$$
$$L(r, \lambda) = -\lambda u(r),$$

so $H(r) - L(r, \lambda) = (1 + \lambda)u(r) + u(x_h - r) > 0$.

Conditional effort For a fixed r,

$$F_e = q\beta e^{-\beta e}\big[H(r) - L(r, \lambda)\big] - \kappa,$$
$$F_{ee} = -q\beta^2 e^{-\beta e}\big[H(r) - L(r, \lambda)\big] < 0.$$

By $s'' < 0$ and $c'' = 0$, conditional effort is therefore unique. When effort is positive, the first-order condition is

$$qs(e) = q - \frac{\kappa}{\beta[H(r) - L(r, \lambda)]}. \tag{20}$$

For each r, define the conditional effort threshold

$$q^0(r; \lambda) := \frac{\kappa}{\beta[H(r) - L(r, \lambda)]}.$$

Since the conditional effort problem is strictly concave, the conditional effort problem has a unique solution

$$e(r; q, \lambda) = \begin{cases} 0, & q \le q^0(r; \lambda), \\ \dfrac{1}{\beta} \log\left(\dfrac{q\beta[H(r) - L(r, \lambda)]}{\kappa}\right), & q > q^0(r; \lambda). \end{cases}$$

Now I can solve the joint problem as a one dimensional problem of r.

The reference point choice. For $r \ge x_h/2$, $H'(r) \le 0$ and $L_r(r, \lambda) < 0$, so $F_r < 0$. Any aspirational optimum therefore lies in $(0, x_h/2)$. On this interval,

$$H'(r) - L_r(r, \lambda) = (1 + \lambda)u'(r) - u'(x_h - r) > 0.$$

Indeed, $r < x_h - r$ and concavity of u imply $u'(r) > u'(x_h - r)$. It follows that

$$\frac{\partial q^0(r; \lambda)}{\partial r} = -\frac{\kappa[H'(r) - L_r(r, \lambda)]}{\beta[H(r) - L(r, \lambda)]^2} < 0.$$

Hence

$$q^0(r; \lambda) \le q^0(0; \lambda) \qquad \text{for every } r \in [0, x_h/2).$$

On this interval, the reference-point first-order condition gives

$$qs(e) = \Pi(r; \lambda) := \frac{\lambda u'(r)}{(1 + \lambda)u'(r) - u'(x_h - r)}$$
$$= \frac{\lambda}{\lambda + 1 - \left(\frac{r+\sigma}{x_h - r + \sigma}\right)^{1-\alpha}}. \tag{21}$$

Since $1-\alpha > 0$, the term

$$\left(\frac{r+\sigma}{x_h - r + \sigma}\right)^{1-\alpha}$$

is strictly increasing in r. Moreover, it is smaller than one on $[0, x_h/2)$, so the denominator is positive and strictly decreasing. Since the numerator is the positive constant λ, it follows that $\Pi_r(r;\lambda) > 0$.

Combining (20) and (21), an interior stationary point satisfies

$$q(r;\lambda) = \Pi(r;\lambda) + \frac{\kappa}{\beta[H(r) - L(r,\lambda)]}. \tag{22}$$

Taking the derivative

$$q_r(r;\lambda) = \Pi_r(r;\lambda) - \frac{\kappa[H'(r) - L_r(r;\lambda)]}{\beta[H(r) - L(r,\lambda)]^2} \tag{23}$$

Since $\Pi_r > 0$, define the finite bound

$$\beta_D(\lambda) := \max_{r\in[0,x_h/2]} \frac{\kappa[H'(r) - L_r(r,\lambda)]}{[H(r) - L(r,\lambda)]^2 \Pi_r(r;\lambda)}. \tag{24}$$

Then $\beta > \beta_D(\lambda)$ implies $q_r(r;\lambda) > 0$.

A higher aspiration requires a higher success probability to satisfy the reference-point FOC, which tends to increase the required productivity. But a higher aspiration also induces more effort (by complementarity), which goes in the opposite direction. When β is large enough, the first effect dominates and $q(r;\lambda)$ is increasing in r.

Second order conditions. First I show that under the assumptions, $D = F_{rr}F_{ee} - F_{re}^2$ is strictly positive. This guarantees that with $F_{ee} < 0$, (e, r) is a strict local maximum. This will also allow me to find the sign of the derivative $\frac{de^*}{d\lambda}$.

Since $F_r = qs(e)[H'(r) - L_r(r,\lambda)] + L_r(r,\lambda)$, and by (21) $L_r(r;\lambda) = -\Pi(r;\lambda)[H'(r) - L_r(r;\lambda)]$, it follows that

$$\begin{aligned}
F_r &= [H'(r) - L_r(r,\lambda)][qs(e) - \Pi(r;\lambda)] \\
F_{rr} &= [H''(r) - L_{rr}(r,\lambda)][qs(e) - \Pi(r;\lambda)] - [H'(r) - L_r(r,\lambda)]\Pi_r(r;\lambda) \\
&= -[H'(r) - L_r(r,\lambda)]\Pi_r(r;\lambda) && \text{by FOC of } r \\
F_{re} &= [H'(r) - L_r(r,\lambda)]qs'(e) \\
&= \frac{\kappa[H'(r) - L_r(r,\lambda)]}{H(r) - L(r,\lambda)} && \text{by FOC of } e \\
F_{ee} &= -q\beta s'(e)[H(r) - L(r,\lambda)] \\
&= -\beta\kappa && \text{by FOC of } e
\end{aligned}$$

Then

$$\begin{aligned} D &= F_{rr}F_{ee} - F_{re}^2 \\ &= (-[H'(r) - L_r(r,\lambda)]\Pi_r(r;\lambda))\,(-\beta\kappa) - \left(\frac{\kappa[H'(r) - L_r(r,\lambda)]}{H(r) - L(r,\lambda)}\right)^2 \\ &= \beta\kappa[H'(r) - L_r(r,\lambda)]\left[\Pi_r(r;\lambda) - \frac{\kappa[H'(r) - L_r(r,\lambda)]}{\beta[H(r) - L(r,\lambda)]^2}\right] \\ &= \beta\kappa[H'(r) - L_r(r,\lambda)]q_r(r;\lambda). \end{aligned} \tag{25}$$

Thus $\beta > \beta_D(\lambda)$ also implies $D > 0$.

Activation and uniqueness. Let

$$\widehat{F}(r;q,\lambda) = F(r, e(r;q,\lambda);q,\lambda).$$

If conditional effort is zero, then $\widehat{F}_r = L_r < 0$ and the agent holds no aspirations $r^* = 0$. If effort is positive, the envelope theorem gives $\widehat{F}_r(r;q,\lambda) = F_r$

$$\begin{aligned} \widehat{F}_r &= [H'(r) - L_r(r,\lambda)][qs(e) - \Pi(r;\lambda)] \\ &= [H'(r) - L_r(r,\lambda)]\left[q - \frac{\kappa}{\beta[H(r) - L(r,\lambda)]} - \Pi(r;\lambda)\right] \\ &= [H'(r) - L_r(r,\lambda)][q - q(r;\lambda)] \end{aligned} \tag{26}$$

If the productivity q exceeds the productivity required to support r, raising the aspiration is profitable. Since $q_r > 0$, $q(0,\lambda)$ is the lowest productivity that can sustain an aspiration. Define the aspiration threshold as

$$\begin{aligned} q^A(\lambda) &= q(0;\lambda) \\ &= \frac{\kappa}{\beta u(x_h)} + \frac{\lambda}{\lambda + 1 - \left(\frac{\sigma}{x_h+\sigma}\right)^{1-\alpha}} \\ &= \frac{\kappa}{\beta u(x_h)} + \Pi(0;\lambda) \\ &= q^0(0,\lambda) + \Pi(0,\lambda) \end{aligned}$$

Since $\Pi(0;\lambda) > 0$ and $q^0(\cdot;\lambda)$ is decreasing,

$$q^A(\lambda) > q^0(0;\lambda) \geq q^0(r;\lambda) \qquad \text{for every } r \in [0, x_h/2). \tag{27}$$

Define

$$\beta_A(\lambda) := \frac{\kappa}{u(x_h)[1 - \Pi(0;\lambda)]}.$$

The condition $\beta > \beta_A(\lambda)$ is equivalent to $q^A(\lambda) < 1$.

- Suppose $q \leq q^A(\lambda)$. Since $q(r;\lambda)$ is strictly increasing, $q(r;\lambda) > q(0;\lambda) = q^A(\lambda) \geq q$ for every $r > 0$. Equation (26) therefore implies $\widehat{F}_r < 0$ whenever conditional effort is positive. If effort is zero,

$\widehat{F}_r = L_r < 0$. So $r^*(q, \lambda) = 0$.

- Now suppose $q > q^A(\lambda)$. By (27), conditional effort is positive on $[0, x_h/2)$. At $r = 0$, $\widehat{F}_r(0; q, \lambda) > 0$ so $r^* > 0$. As $r \uparrow x_h/2$, $\Pi(r; \lambda) \to 1$, and continuity of $H - L$ gives

$$\lim_{r \uparrow x_h/2} q(r; \lambda) = 1 + \frac{\kappa}{\beta[H(x_h/2) - L(x_h/2, \lambda)]} > 1.$$

  By continuity of $q(\cdot; \lambda)$, there exists a solution $r^*(q, \lambda) \in (0, x_h/2)$ to $q = q(r^*; \lambda)$. Since $q_r > 0$, equation (26) implies that $\widehat{F}_r$ changes sign at most once, from positive to negative. Hence $r^*(q, \lambda)$ is the unique aspirational optimum.

Finally, the aspirational reference point satisfies $q = q(r^*; \lambda)$. Differentiating with respect to q gives

$$\frac{dr^*}{dq} = \frac{1}{q_r(r^*; \lambda)} > 0. \tag{28}$$

Since $q^A(\lambda) = q(0; \lambda)$ and $q(r; \lambda)$ is continuous and strictly increasing, $r^*(q, \lambda) \downarrow 0$ as $q \downarrow q^A(\lambda)$.

Loss sensitivity and the reference point. I next show that the aspirational reference point decreases with loss sensitivity. Since $q = q(r^*; \lambda)$, differentiating with respect to λ gives

$$\frac{dr^*}{d\lambda} = -\frac{q_\lambda(r^*; \lambda)}{q_r(r^*; \lambda)}.$$

Since $q_r > 0$, I need to show that $q_\lambda > 0$.

From (22), and since $\partial_\lambda[H(r) - L(r, \lambda)] = u(r)$,

$$q_\lambda = \Pi_\lambda - \frac{\kappa u(r)}{\beta[H(r) - L(r, \lambda)]^2}.$$

At an interior optimum,

$$\frac{\kappa}{\beta[H(r) - L(r, \lambda)]} = q - \Pi(r; \lambda).$$

Hence

$$\begin{aligned} q_\lambda &= \Pi_\lambda - [q - \Pi(r; \lambda)] \frac{u(r)}{H(r) - L(r, \lambda)} \\ &\geq \Pi_\lambda - [1 - \Pi(r; \lambda)] \frac{u(r)}{H(r) - L(r, \lambda)} \quad \text{since } q \leq 1. \end{aligned}$$

From the definition of Π,

$$\Pi(r; \lambda) = \frac{\lambda u'(r)}{(1 + \lambda)u'(r) - u'(x_h - r)}.$$

Differentiating with respect to λ and using the expression for $1 - \Pi(r; \lambda)$ gives

$$\begin{aligned} \Pi_\lambda &= \frac{u'(r)\big[(1 + \lambda)u'(r) - u'(x_h - r)\big] - \lambda[u'(r)]^2}{\big[(1 + \lambda)u'(r) - u'(x_h - r)\big]^2} \\ &= \frac{u'(r)\big[u'(r) - u'(x_h - r)\big]}{\big[(1 + \lambda)u'(r) - u'(x_h - r)\big]^2} \\ &= [1 - \Pi(r; \lambda)] \frac{u'(r)}{(1 + \lambda)u'(r) - u'(x_h - r)} \end{aligned}$$

where the last step comes from factorizing out $1 - \Pi(r;\lambda) = \frac{u'(r)-u'(x_h-r)}{(1+\lambda)u'(r)-u'(x_h-r)}$.

Therefore

$$q_\lambda \geq [1 - \Pi(r;\lambda)] \left[\frac{u'(r)}{(1+\lambda)u'(r) - u'(x_h - r)} - \frac{u(r)}{H(r) - L(r,\lambda)} \right].$$

The term in brackets is positive since

$$\begin{aligned} &u'(r)[H(r) - L(r,\lambda)] - u(r)\big[(1+\lambda)u'(r) - u'(x_h - r)\big] \\ &= u'(r)u(x_h - r) + u(r)u'(x_h - r) > 0. \end{aligned}$$

Thus $q_\lambda > 0$, and therefore $\frac{dr^*}{d\lambda} < 0$.

Effect of loss sensitivity on effort. Finally, I show where loss sensitivity changes from reducing effort to increasing it. Recall that, along the stable aspirational branch,

$$\operatorname{sgn}\left(\frac{de^*}{d\lambda}\right) = \operatorname{sgn}[R(r^*,\lambda) - 1].$$

First, I show that $R(r,\lambda)$ crosses one at a unique reference point $r^E(\lambda)$. Second, I use $q_r > 0$ to translate this reference point cutoff into a unique productivity cutoff $q^E(\lambda)$.

Step 1. For shifted-power utility,

$$\Delta\rho(r) = \frac{(1-\alpha)\left(\frac{r+\sigma}{x_h - r + \sigma}\right)^{1-\alpha}\left(1 + \frac{r+\sigma}{x_h - r + \sigma}\right)}{(r+\sigma)\left[1 - \left(\frac{r+\sigma}{x_h - r + \sigma}\right)^{1-\alpha}\right]}. \tag{29}$$

For $\alpha \neq 0$, the motivation index is

$$R(r,\lambda) = \Pi(r;\lambda)(1-\alpha)\frac{1 - \left(\frac{\sigma}{r+\sigma}\right)^{\alpha}}{\alpha}\frac{\left(\frac{r+\sigma}{x_h - r + \sigma}\right)^{1-\alpha}\left(1 + \frac{r+\sigma}{x_h - r + \sigma}\right)}{1 - \left(\frac{r+\sigma}{x_h - r + \sigma}\right)^{1-\alpha}}. \tag{30}$$

When $\alpha = 0$, replace the first fraction after $1 - \alpha$ with $\log\left(\frac{r+\sigma}{\sigma}\right)$.

The factors in (30) are strictly increasing on $(0, x_h/2)$.

$$\frac{d}{dr}\left[\frac{1 - \left(\frac{\sigma}{r+\sigma}\right)^{\alpha}}{\alpha}\right] = \frac{\left(\frac{\sigma}{r+\sigma}\right)^{\alpha}}{r+\sigma} > 0,$$

and the last fraction has an increasing numerator and a decreasing positive denominator. Together with $\Pi_r > 0$, this gives $R_r(r,\lambda) > 0$. Since $\lim_{r\downarrow 0} R(r,\lambda) = 0$ and $\lim_{r\uparrow x_h/2} R(r,\lambda) = +\infty$, there is therefore a unique $r^E(\lambda) \in (0, x_h/2)$ satisfying $R(r^E(\lambda),\lambda) = 1$.

Step 2. Now I show that this translates into a unique productivity between 0 and 1, at which $R = 1$.

$$q^E(\lambda) := q(r^E(\lambda);\lambda) = \Pi(r^E(\lambda);\lambda) + \frac{\kappa}{\beta[H(r^E(\lambda)) - L(r^E(\lambda),\lambda)]}.$$

Recall that $q^A(\lambda) = q(0;\lambda)$. Since $r^E(\lambda) > 0$ and $q_r > 0$, then $q^E(\lambda) > q^A(\lambda)$.

Define

$$\beta_E(\lambda) := \frac{\kappa}{[H(r^E(\lambda)) - L(r^E(\lambda), \lambda)][1 - \Pi(r^E(\lambda); \lambda)]},$$
$$\bar{\beta}(\lambda) := \max\{\beta_A(\lambda), \beta_D(\lambda), \beta_E(\lambda)\}.$$

Then $\beta > \bar{\beta}(\lambda)$ implies

$$0 < q^A(\lambda) < q^E(\lambda) < 1.$$

For $q < q^A(\lambda)$, $r^* = 0$, so effort is independent of λ. For $q^A(\lambda) < q < q^E(\lambda)$, $r^* < r^E(\lambda)$, and the general sign result gives $de^*/d\lambda < 0$. At $q = q^E(\lambda)$, $R(r^*, \lambda) = 1$, so $de^*/d\lambda = 0$. For $q > q^E(\lambda)$, $R(r^*, \lambda) > 1$, so $de^*/d\lambda > 0$.

■

A.8.3 Proof of Corollary 5

Proof. The thresholds are

$$q^A(\lambda) = \Pi(0; \lambda) + \frac{\kappa}{\beta u(x_h)},$$
$$q^E(\lambda) = \Pi(r^E; \lambda) + \frac{\kappa}{\beta[H(r^E) - L(r^E, \lambda)]}.$$

The threshold $r^E(\lambda)$ is defined by $R(r^E, \lambda) = 1$. Since $R(r, \lambda)$ does not depend on β or κ, neither does $r^E(\lambda)$. So it follows directly that both $q^A(\lambda)$ and $q^E(\lambda)$ decrease with β and increase with κ.

Finally,

$$\frac{dq^A(\lambda)}{d\lambda} = \frac{1 - \left(\frac{\sigma}{x_h+\sigma}\right)^{1-\alpha}}{\left[\lambda + 1 - \left(\frac{\sigma}{x_h+\sigma}\right)^{1-\alpha}\right]^2} > 0.$$

■